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SynNeura: Event-driven liquid-spiking dynamics for weakly supervised multimodal temporal alignment

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ABSTRACT

Multimodal temporal alignment is a critical task for applications such as audiovisual understanding, lip reading, and instruction following. However, in weakly supervised settings, challenges like asynchronous sampling, irregular event triggers, and coarse labels hinder precise cross-modal alignment. Frame-based methods rely on fixed time grids, leading to redundant computation in sparse-event scenarios and reducing event-level precision. Differentiable time-warping methods typically require high-resolution inputs, resulting in high computational costs and sensitivity to numerical parameters. To address these challenges, we propose SynNeura, an event-driven continuous-time liquid-spiking neural framework for fine-grained alignment under weak supervision. SynNeura models alignment as a continuous-time latent-state process, analytically propagating states between events and updating only when spikes occur. This results in computational complexity that scales with the number of events rather than the sequence length. SynNeura introduces a piecewise-analytic update using matrix exponentials and trace variables, along with a hierarchical contrastive alignment objective at spike, trajectory, and state levels, enhancing robustness and consistency. Experiments with the AVE, LRS2, and YouCook2 datasets show SynNeura consistently outperforms frame-based and continuous-time baselines in alignment accuracy, temporal consistency, and efficiency. SynNeura achieves 0.291 MAE and 0.713 CAS on AVE, 0.648 TC on LRS2, and 0.829/0.794 EP/ER on YouCook2, with an overall score of 0.738. These results show SynNeura is a scalable, interpretable, and efficient solution for event-driven multimodal temporal alignment.

1. Introduction

Multimodal temporal alignment aims to establish semantically consistent temporal correspondences across heterogeneous multimodal streams with diverse temporal structures, and constitutes a foundational capability for applications such as audiovisual understanding, lip reading, instruction grounding, and robotic learning (Baltrušaitis et al., 2019; Song et al., 2013; Zadeh et al., 2018). In real-world scenarios, multimodal signals are inherently asynchronous, event-sparse, and irregularly sampled, often accompanied by uncertain event boundaries and incomplete temporal structures (Kidger et al., 2020; Tsai et al., 2020). These properties pose substantial challenges to modeling cross-modal temporal consistency. Under weak supervision, where only coarse-grained labels or pseudo-alignment cues are available, learning reliable event-level alignment in the presence of complex temporal dynamics becomes particularly difficult.

Existing temporal alignment methods can be broadly categorized into two paradigms. The first paradigm consists of implicit alignment approaches based on discretized temporal correlation structures (Elsayed et al., 2019; Lim et al., 2021; Wu et al., 2021), including hierarchical sequence modeling, autocorrelation mechanisms, and temporal attention. While effective in capturing long-range dependencies, these methods discretize time a priori and infer alignment mainly through implicit correlations, making it challenging to model temporal correspondence dominated by sparse key events. Moreover, their computational and memory costs often grow significantly with sequence length, limiting scalability to long-duration sequences. The second paradigm includes explicit matching approaches based on differentiable temporal deformation, such as Soft-DTW (Cuturi & Blondel, 2017). These methods mitigate discontinuities in discrete matching by learning continuous warping paths but typically rely on dense temporal sampling, which rapidly increases complexity with higher temporal resolution and reduces practicality in sparse and asynchronous multimodal settings.

Recent advances in Spiking Neural Networks(SNNs) (Deng et al., 2022; Fang et al., 2021; Roy et al., 2019; Tavanaei et al., 2019) and continuous-time dynamical models, exemplified by Liquid Neural

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Networks(LNNs) (Hasani et al., 2021, 2022), provide promising avenues for event-centric temporal modeling. SNNs operate on spike events as fundamental computational units, enabling event-triggered state updates and fine-grained temporal correspondence modeling. Continuous-time dynamical systems explicitly characterize the evolution of latent states over time, offering a principled coupling between temporal dynamics and cross-modal representations. Nevertheless, most existing spiking models rely on simplified integrate-and-fire dynamics or are unfolded over discrete time steps, limiting temporal continuity and expressive power. Conversely, continuous-time models often depend on numerical integration over dense temporal grids, making their computational cost and numerical stability highly sensitive to step-size selection. As a result, current approaches struggle to simultaneously achieve strong expressiveness, efficiency, and scalability for weakly supervised multimodal temporal alignment.

It is worth noting that continuous-time models and spiking neural networks provide effective tools for irregular sequence modeling and event-driven computation, respectively. However, directly cascading an SNN encoder with a Neural CDE/ODE solver is insufficient to fully address event-level correspondence in weakly supervised multimodal temporal alignment. The key reason is that spike events should not be treated merely as passive inputs fed into a continuous-time solver; instead, they should actively determine when the hidden state is analytically propagated and when discontinuous jump updates are triggered. In other words, an ideal model for weakly supervised multimodal temporal alignment should be driven by cross-modal event arrivals, rather than by predefined temporal grids or dense numerical integration, thereby better accommodating long-sequence, low-sampling-rate, and event-sparse scenarios.

To overcome these limitations, we propose SynNeura. This event-driven liquid spiking neural dynamical model integrates the high-temporal-precision event representation of the Spike Response Model(SRM) with the adaptive continuous-time modeling capacity of liquid neural dynamics. SynNeura formulates multimodal temporal alignment as a unified continuous-time spiking process: latent neural states evolve analytically between sparse events and are updated only upon spike arrivals. Inter-event evolution admits a piecewise exact solution via matrix exponentials and trace variables, enabling numerically stable inference with computational complexity that scales linearly with the number of events rather than the number of discretized time steps. This property is particularly critical for long-duration, low-sampling-rate, and highly asynchronous multimodal sequences.

Under weak supervision, we further design a hierarchical contrastive alignment objective that imposes multi-scale constraints at the spike, trajectory, and state levels. This formulation jointly captures local event correspondences and global dynamical trajectory structures, improving robustness and interpretability in scenarios with limited supervision and pronounced cross-modal temporal discrepancies.

The main contributions of this work are summarized as follows:

- We propose SynNeura, an event-driven continuous-time framework for multimodal temporal alignment. Rather than simply cascading an SNN encoder with a Neural CDE/ODE solver, SynNeura formulates weakly supervised multimodal alignment as a continuous-time spiking dynamical process. By performing closed-form analytical propagation between events and sparse nonlinear state updates upon event arrivals, SynNeura reduces redundant computation in silent regions and improves event-level temporal alignment under weak supervision, low-sampling-rate conditions, and long-sequence scenarios.
- We derive an event-driven liquid-state inference mechanism based on matrix exponentials and trace variables. This mechanism performs closed-form analytical propagation during inter-event intervals and executes major nonlinear state updates only upon event arrival. As a result, it avoids high-frequency numerical integration over dense temporal grids, reduces sensitivity to step-size selection, and improves numerical stability and inference efficiency in long-sequence scenarios.
- We construct a three-level contrastive alignment objective at the spike, trajectory, and state levels. These objectives respectively constrain local event correspondence, global trajectory consistency, and continuous-state stability, thereby promoting semantically consistent cross-modal temporal alignment across multiple time scales under weak supervision.
- We evaluate SynNeura on the AVE (Tian et al., 2018), LRS2 (Chung et al., 2017), and YouCook2 (Zhou et al., 2018) datasets. Our experiments, ablation studies, and complexity analyses show that our framework delivers strong alignment accuracy, robustness, and inference efficiency for weakly supervised multimodal temporal alignment. These results confirm the impact of our key components: event-based encoding, event-driven liquid-state propagation, and the multi-level contrastive alignment objective.

The remainder of this paper is organized as follows. We first formalize the multimodal temporal alignment problem and introduce the proposed spike-response-based dynamical formulation. Next, we derive the alignment-constrained liquid-state evolution and present the hierarchical contrastive objectives. Finally, we analyze the computational complexity and stability properties of SynNeura and report comprehensive experimental results.

2. Related works

We categorize prior research into three lines: multimodal temporal correspondence learning, event-driven and continuous-time neural modeling, and contrastive learning under weak temporal supervision. Our work bridges these areas by jointly leveraging event-triggered computation and continuous-time dynamics to model fine-grained temporal correspondences, making it particularly suitable for weak supervision and strong cross-modal asynchrony.

2.1. Multimodal temporal correspondence learning

Multimodal temporal alignment aims to identify event onsets, durations, and possible temporal shifts across modalities. Early video–language grounding works, such as MCN and DiDeMo, established task formulations and benchmark protocols (Hendricks et al., 2017). Subsequent studies strengthened alignment modeling through explicit interval reasoning (2D-TAN) (Zhang et al., 2020b), end-to-end boundary regression (VSLNet) (Zhang et al., 2020a), and multi-granular representations for long videos (QVHighlights) (Lei et al., 2021). UniVTG and ProTeGe further improved sensitivity to boundary cues and semantic transitions (Lin et al., 2023; Wang et al., 2023b). More recently, DETR-style temporal localization has incorporated temporal sampling and set-based prediction to enhance alignment quality (Wang et al., 2023a).

Most of these methods, however, operate on regularly sampled, discretized video sequences. In sparse-event or strongly asynchronous scenarios, such discretization can introduce substantial redundancy and may obscure fine-grained event correspondences. To reduce reliance on dense frames under weak supervision, recent work has explored weakly supervised temporal grounding and moment retrieval, including query-aware multi-scale proposal modeling for temporal sentence grounding (Zhou et al., 2024) and weakly supervised content-based moment retrieval with improved video representations (Huo et al., 2023). In addition, weakly supervised spatial–temporal grounding with minimal annotation (e.g., spatial–temporal supervision on a single frame) has been investigated to further alleviate labelling costs while maintaining localization capability (Luo et al., 2025).

In addition, complementary work has explored alternative alignment strategies. For example, some studies model temporal alignment in the compressed video domain to reduce reliance on explicit frame decoding (Fang et al., 2023b), and in cross-modal settings researchers leverage cross-modal consistency as a self-supervision signal—such as audio–visual and temporal cycle consistency—to improve alignment without dense annotations (Arandjelović & Zisserman, 2017; Dwibedi et al., 2019; Morgado et al., 2021). Notably, Zhang et al. (2024) proposed a center-enhanced captioning model that learns shared latent multimodal representations to strengthen semantic alignment. Despite these advances, a unified framework that treats events as fundamental units, efficiently models continuous-time dependencies, and remains effective under weak supervision is still lacking.

2.2. Event-driven and continuous-time neural modeling

Event-driven sensors (e.g., event cameras) generate sparse and asynchronous event streams, motivating continuous-time modeling rather than frame-based conversion (Gallego et al., 2022). Recent event-based vision methods integrated with deep sequential modeling approaches, such as RNNs and Transformers, have achieved promising performance across a wide range of tasks, including recognition, detection, person re-identification, video restoration, and deraining (Cao et al., 2026; Fu et al., 2024; Gehrig & Scaramuzza, 2023). At the same time, state-space formulations provide an efficient framework for modelling irregular event dynamics (Zubić et al., 2024).

Continuous-time neural modeling offers a principled dynamical systems perspective for nonuniformly sampled temporal signals. Neural ODEs represent latent dynamics as continuous-time processes (Chen et al., 2018), and Neural CDEs further generalize this paradigm to irregularly observed sequences (Morrill et al., 2021). Liquid neural networks enhance adaptability via state-dependent time constants (Lechner et al., 2020), whereas closed-form continuous-time networks improve inference efficiency through analytical state propagation (Rubanova et al., 2019).

Complementary to these formulations, SNNs naturally support event-triggered computation through spike-driven state updates. Recent advances integrate attention mechanisms into spiking models and develop efficient training strategies for practical deployment (Fang et al., 2023a; Shi et al., 2024; Wang et al., 2023c; Zhou et al., 2023).

Moreover, event-driven video restoration methods have further integrated spiking computation with convolutional architectures, demonstrating the suitability of SNN-like models for efficient dynamic vision restoration (Cao et al., 2025). Under limited-annotation settings, biologically inspired active learning mechanisms have also been explored to reduce annotation requirements (Zhan et al., 2023), which aligns well with our objective of weakly supervised cross-modal alignment.

Despite these advances, most continuous-time and liquid models remain unimodal and lack explicit cross-modal semantic guidance. In contrast, spiking models often lack end-to-end coupling between event-driven dynamics and alignment-oriented objectives. Thus, unifying continuous-time evolution, event-triggered computation, and cross-modal alignment constraints in asynchronous event-driven scenarios remains an open challenge.

2.3. Contrastive learning under weak temporal alignment conditions

Contrastive learning has emerged as a central paradigm for cross-modal representation learning, encouraging semantically consistent embeddings from different modalities to align in a shared latent space by constructing positive and negative pairs. Large-scale vision–language contrastive pretraining models, exemplified by CLIP, have demonstrated strong generalization and have further advanced video–text representation learning (Radford et al., 2021). Subsequent works, such as VideoCLIP and Frozen-in-Time, extend contrastive objectives to sequential video data by incorporating temporal structure via segment-level alignment (Bain et al., 2021; Xu et al., 2021).

In real-world multimodal scenarios, however, temporal correspondence across modalities is often only weakly supervised and frequently subject to systematic misalignment. To mitigate noisy supervision induced by narration–visual asynchrony, MIL-NCE introduces a multiple-instance contrastive objective that improves robustness under temporal mismatch (Miech et al., 2020). ReLoCLNet further enhances localization performance under weak temporal alignment by refining the construction of positive and negative samples (Zhang et al., 2021). Nevertheless, most existing contrastive frameworks still rely on frames or fixed-length clips as the basic alignment units, implicitly assuming regular temporal sampling. Such discretization becomes restrictive when alignment cues are sparse, asynchronous, and event-triggered, limiting the expressiveness and interpretability of contrastive objectives in fine-grained temporal alignment settings. Moreover, current weakly supervised approaches remain largely grounded in discrete temporal grids, making them difficult to integrate naturally with event-driven computation and continuous-time dynamical evolution.

Overall, although notable progress has been made in multimodal temporal alignment, continuous-time modelling, event-driven computation, and weakly supervised contrastive learning, these directions are often developed in isolation. This gap motivates SynNeura, which unifies event-triggered computation, continuous-time dynamics, and hierarchical contrastive learning objectives to enable weakly supervised multimodal temporal alignment in asynchronous event-driven scenarios.

3. Methods

3.1. Problem formulation

Multimodal temporal alignment aims to establish semantically consistent temporal correspondences across asynchronous and multi-source sequential data. In real-world scenarios, different modalities often exhibit discrepancies in sampling rates and systematic delays or temporal jitter. Moreover, the training stage typically lacks frame-level precise annotations, providing only coarse video-level or segment-level labels or weak alignment cues. Given two irregularly sampled modality sequences $X = \left\{\left(t_i^x, x_i\right)\right\}_{i=1}^{N_x}$ and $Y = \left\{\left(t_i^y, y_i\right)\right\}_{i=1}^{N_y}$.

Our goal is to learn model parameters θ from a weakly supervised set S, such that cross-modal semantic correspondences are temporally aligned. The overall optimization objective is formulated as $\min_\theta \sum_{(x,y)} \left(\mathcal{L}_{align}(x, y; \theta) + \lambda R(\theta)\right)$.

Where $\mathcal{L}_{align}$ consists of event-level alignment, trajectory-level consistency, and dynamical regularization terms (see Section 3.4), and $R(\theta)$ denotes a regularization term. To simultaneously achieve alignment accuracy and computational efficiency under long sequences and sparse event conditions, we propose an event-driven continuous-time liquid spiking dynamical framework, termed SynNeura (Fig. 1).

3.2. SRM-based event encoding for multimodal alignment

Multimodal temporal alignment faces the challenge of varying sampling rates, asynchronous delays, and jitter across modalities such as speech, vision, and text, making precise alignment difficult through a unified frame rate. In scenarios with sparse or asynchronous events, traditional frame- or window-based alignment methods fail to capture key moments of cross-modal interaction. To address this, we introduce the SRM as the event-level encoding front-end, converting multimodal continuous signals into sparse, time-precise event streams. This event-driven approach avoids frame redundancy and high latency, enabling more effective modelling of high-resolution temporal dependencies between asynchronous modalities.

In SRM, the membrane potential of each neuron is determined by both the synaptic response to input events and the inhibitory effects from previous spikes. The potential of the i-th neuron is given by:

$$V_i(t) = \sum_m \sum_j \sum_{t_{j,k}^{(m)} \in S_j^{(m)}} w_{i,j}^{(m)} \, \varepsilon^{(m)}\left(t - t_{j,k}^{(m)}\right) + \sum_{t_{i,\ell} \in S_i} \eta\left(t - t_{i,\ell}\right) \tag{1}$$

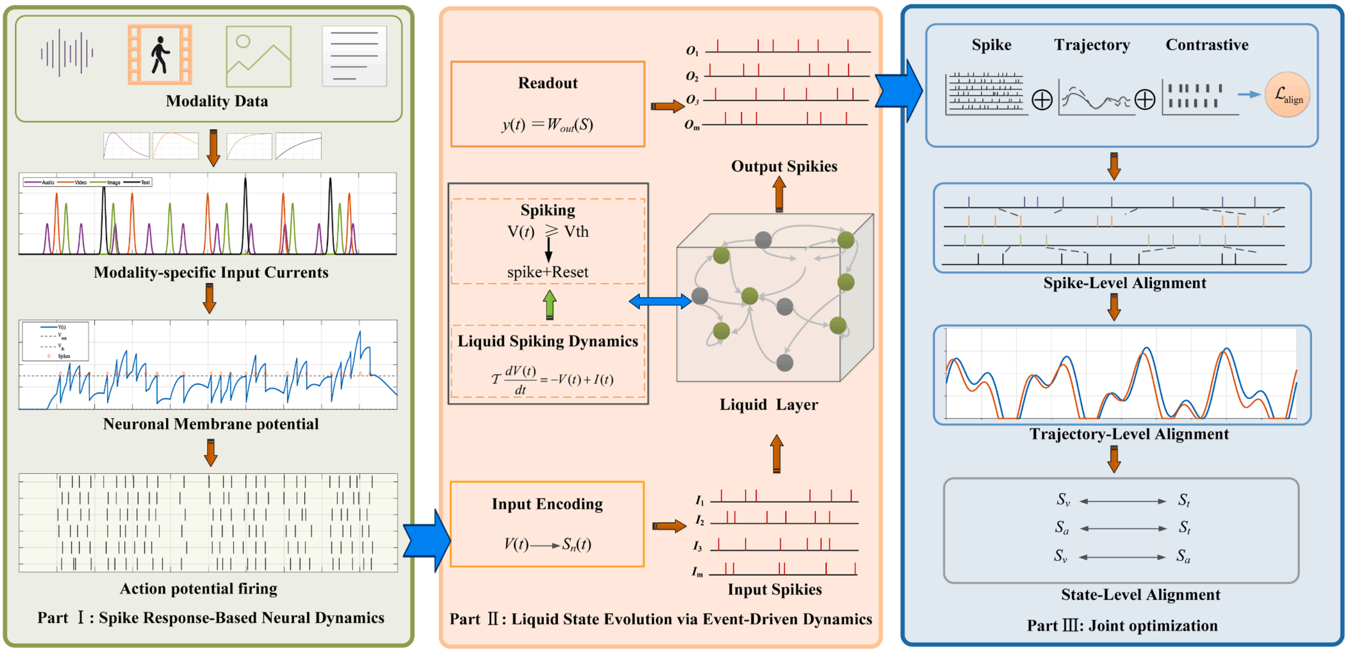


Fig. 1. Overall architecture of the event-driven liquid spiking neural dynamics model (SynNeura).

where m denotes the modality, $S_j^{(m)}$ represents the set of event times for the precursor neuron j in modality m, S_i is the set of past spike times for neuron i, and $w_{ij}^{(m)}$ is the synaptic weight. The functions $\epsilon^{(m)}(\cdot)$ and $\eta(\cdot)$ are the input response and spike suppression kernels, respectively.

When the membrane potential exceeds a threshold θ_i, the neuron triggers a spike: $s_i(t) = H(V_i(t) - \theta_i)$. where $H(\cdot)$ is the Heaviside step function. During training, a differentiable approximation of $H(\cdot)$ is used for end-to-end gradient propagation (see Section 3.4).

To ensure the temporal causality of the input response and to characterize the membrane-potential dynamics in which the response rapidly rises after an input event and subsequently decays over time, we model the input response kernel as a causal bi-exponential function: $\epsilon^{(m)}(t) = A_m\left(e^{-t/\tau_{d,m}} - e^{-t/\tau_{r,m}}\right) \cdot H(t)$, where $\tau_{r,m}$ and $\tau_{d,m}$ denote the rise and decay time constants of the m-th component, respectively, which characterize the temporal scales of the rising and decaying phases of the membrane-potential response. The Heaviside function $H(t)$ ensures that the response is activated only after an event occurs. Compared with a single-exponential kernel, the bi-exponential kernel can simultaneously describe both the rising and decaying phases of the response, thereby more accurately capturing the transient membrane-potential variations of event-driven neurons and improving the interpretability of cross-modal temporal response modelling.

The inhibitory kernel $\eta(t)$ is introduced to characterize the recovery process of a neuron after it emits a spike. After firing, a neuron typically enters a transient refractory or inhibitory state, during which its membrane potential is temporarily suppressed and then gradually recovers. Without explicitly modelling historical spike emissions, the model may generate excessively frequent repeated activations within a short time interval, thereby leading to unstable temporal responses. Therefore, we formulate the inhibitory kernel as a causal exponential decay function: $\eta(t) = -A_\eta \exp(-\beta_\eta t) H(t)$. Here, A_η denotes the inhibition strength, which controls the magnitude by which the membrane potential is suppressed after firing; β_η denotes the exponential decay rate, which determines how rapidly the inhibitory effect vanishes over time; and $H(t)$ is the Heaviside function, ensuring that the inhibitory effect is activated only after the neuron emits a spike. Owing to its negative sign, this term suppresses the membrane potential.

It should be noted that β_η is not a manually predefined fixed hyperparameter, but is instead treated as a learnable parameter and jointly optimised with the other model parameters during training. This design is motivated by the fact that different modalities, neurons, and temporal sequences often exhibit distinct temporal sparsity patterns and triggering frequencies. To ensure that the decay rate of the inhibitory kernel remains positive, we adopt the following parameterization: $\beta_\eta = \mathrm{softplus}\left(\tilde{\beta}_\eta\right)$. Here, $\tilde{\beta}_\eta$ is an unconstrained learnable variable. This design enables the model to adaptively adjust the inhibitory recovery timescale following neuronal firing based on the data. When β_η is large, the inhibitory term decays more rapidly, allowing the neuron to recover its responsiveness within a shorter time interval; when β_η is small, the inhibitory effect persists for a longer duration, thereby imposing a stronger constraint on high-frequency repeated firing. By learning β_η, the model can adapt to the temporal sparsity, triggering frequency, and cross-modal response discrepancies across different modal event streams, thereby enhancing its ability to model complex temporal event patterns and characterize the post-spike recovery process.

In summary, the exponential-decay inhibitory kernel does four things: it characterizes the neuronal recovery process after spike emission, suppresses short-term repeated triggering, stabilizes membrane-potential dynamics, and enhances cross-modal temporal modelling at the model level. In contrast to a fixed decay rate, the learnable β_η improves the model's adaptability to event patterns across different temporal scales. This enables the SRM neuron to automatically establish a history-dependent inhibitory mechanism that more closely matches the underlying data during training.

To achieve unified modelling of multimodal event streams, we stack the events from all modalities and express them in vectorized form:

$$V(t) = \sum_m W_m(\epsilon^{(m)} * s_m)(t) + W_{fb}(\eta * s)(t) + b \tag{2}$$

where $s_m(t)$ is the spike event stream for modality m, $s(t)$ is the output spike stream, and W_m, W_{fb} are the weight matrices for modality-specific and feedback pathways. Here, * denotes temporal convolution, and b is the bias term. This representation maps events from different modalities into a unified neural state $V(t)$, providing an event-granularity input interface for alignment and modelling.

SRM follows an event-driven paradigm: during periods without new events, the state decays exponentially; when an event occurs, the state is updated instantaneously. For instance, in a first-order leaky model, if no event occurs between t_e and t_{e+1}, the membrane potential satisfies:

$$\frac{dV_i(t)}{dt} = -\frac{1}{\tau} V_i(t) \tag{3}$$

with the solution: $V_i(t) = V_i(t_e)e^{-(t-t_e)/\tau_v}$, the time constant is τ.

Upon the arrival of a new event, the potential is immediately updated according to the aforementioned convolution formula. Thus, the SRM captures continuous-time dynamics with smooth inter-event decay and event-triggered jump updates, naturally using events as the smallest units for alignment.

In summary, the SRM module encodes continuous multimodal signals into sparse, time-precise event sequences and outputs a unified event-driven state representation, $V(t)$. This provides an efficient and alignment-friendly input mechanism for subsequent liquid neural models (Section 3.3), significantly enhancing the ability and efficiency of asynchronous cross-modal modelling.

3.3. Event-driven liquid-state dynamics with alignment constraints

In Section 3.2, we used the SRM to encode multimodal continuous signals into temporally precise spike-event streams. The resulting membrane potential $V(t)$ serves as an event-driven input representation for subsequent continuous-time state evolution. Based on this, we introduce a continuous-time liquid state $h(t)$ as a cross-modal "temporal memory pool" that integrates event-level inputs and captures long-range dependencies in the continuous-time domain. Unlike traditional dynamical systems that integrate over fixed time steps, we employ an event-driven, piecewise-analytic propagation: the state propagates in closed form between events and is updated only at event times, thereby maintaining efficiency and stability even for long sequences and sparse events.

SynNeura differs fundamentally from the standard pipeline that combines a Neural CDE with an SNN encoder, particularly in how spike events are utilized and how continuous-time states are updated. In a conventional architecture where an SNN encoder is coupled with a Neural CDE, spike outputs are typically treated as encoded observations or control signals for a continuous-time solver. At the same time, the latent state is still primarily evolved through numerical integration over predefined temporal discretizations or interpolated paths. In contrast, SynNeura treats spike events as the basic computational units: the arrival time of each event directly determines when the liquid state is updated, while the state between adjacent events is propagated in closed form. Therefore, spike events are not merely input features for a downstream continuous-time model, but explicitly govern the propagation and jump-update process of the liquid state.

This design establishes an event-driven inference mechanism, in which the dominant computational cost scales with the number of events rather than the length of a dense temporal grid. Meanwhile, cross-modal alignment feedback can be directly injected into the state evolution process via event-triggered updates, rather than imposed solely as a posterior sequence-level loss. Thus, SynNeura is not a simple cascade of an SNN encoder and a Neural CDE module, but an event-driven continuous-time alignment framework that couples spike-event encoding, closed-form liquid-state propagation, and hierarchical alignment constraints.

Continuous-Time Liquid Dynamics and Alignment Feedback Terms

For the i-th liquid unit, the state evolution is defined as:

$$\frac{dh_i(t)}{dt} = -\alpha h_i(t) + V_i(t) + \sum_k U_{ik} h_k(t) + g_i(t) \tag{4}$$

Here, $V_i(t)$ denotes the SRM-derived event-driven input representation. It is obtained from the spike-response membrane-potential dynamics and serves as the external driving input for the state evolution of the i-th liquid unit. The parameter a denotes the leakage coefficient. U_{ik} represents the coupling weight within the liquid layer. $g_i(t)$ denotes the feedback term induced by the alignment constraint, representing cross-modal consistency feedback. Specifically, this feedback is derived by applying either a subsequent energy model or a competitive mechanism, which quantitatively formalises the alignment constraint. In this way, alignment information is explicitly injected into the state evolution process.

Vectorizing Eq. (4), let $h(t) \in \mathbb{R}^n$, $V(t) \in \mathbb{R}^n$, and $g(t) \in \mathbb{R}^n$, we obtain:

$$\frac{dh(t)}{dt} = Ah(t) + Bu(t), \quad u(t) = V(t) + g(t) \tag{5}$$

where: $A = \frac{1}{\tau}(U - \alpha I), \quad B = \frac{1}{\tau} I$.

The alignment feedback term $g_i(t)$ is generated by a learnable cross-modal alignment feedback module, rather than relying on external event-level or frame-level alignment annotations. For an event i in modality A, we first construct a candidate set $N(i)$ from modality B according to temporal proximity. The soft alignment weight between event i and candidate event j is then computed as $\alpha_{ij}^{A\leftarrow B} \frac{\exp\left(\mathrm{sim}(e_i^A, e_j^B)/\tau_g\right)}{\sum_{k\in N(i)} \exp\left(\mathrm{sim}(e_i^A, e_k^B)/\tau_g\right)}$, where e_i^A and e_j^B denote event-level embeddings, $\mathrm{sim}(\cdot,\cdot)$ denotes cosine similarity, and τ_g is the temperature parameter.

Based on the soft alignment weights, the feedback term is generated by aggregating cross-modal residuals and projecting them into the liquid state space: $g_i^A(t) = W_g\left(\sum_{j\in N(i)} \alpha_{ij}^{A\leftarrow B}\left(e_j^B - e_i^A\right)\right)$, where W_g is a learnable projection matrix. The feedback term for modality B is computed symmetrically.

Therefore, we formulate the similarity-weighted cross-modal alignment residual as the feedback term $g_i(t)$ and explicitly inject it into the liquid-state dynamics, thereby enabling the state evolution to adaptively adjust based on the current event-level alignment confidence. In this way, the model can incorporate cross-modal consistency constraints in real time during continuous-time evolution, without requiring additional fine-grained alignment annotations. Compared with methods that rely on external alignment labels or fixed alignment rules, the proposed mechanism is end-to-end learnable, event-level adaptive, computationally efficient, and highly interpretable.

It should be clarified that setting $g(t) = 0$ is used only to illustrate the basic closed-form propagation of the liquid dynamics. In the actual implementation of SynNeura, $g(t)$ is always retained and computed from cross-modal event-level alignment residuals. This means the alignment constraint is not just imposed later as a posterior loss. Instead, it is directly injected into the continuous-time state-evolution process. As a result, the liquid state is jointly driven by both sparse multimodal events and alignment feedback. This ensures consistency between the model motivation and its mathematical formulation.

Event-Driven Piecewise Analytic Propagation

In an event-driven system, state evolution is affected not only by the input at the current time instant but also by previous input states. Between any two adjacent event times t_e and t_{e+1}, the input signal can typically be regarded as constant or slowly varying. In this case, the state evolution can be derived through a closed-form solution. Specifically, the state evolution within an inter-event interval can be written as

$$h(t_{e+1}^-) = e^{A\Delta t_e} h(t_e^+) + \int_{t_e}^{t_{e+1}} e^{A(t_{e+1}-s)} Bu(s), ds, \tag{6}$$

where $\Delta t_e = t_{e+1} - t_e$ denotes the time interval between two adjacent events, $h(t_{e+1}^-)$ is the state immediately before event $e+1$, and $h(t_e^+)$ is the state immediately after event e.

It should be emphasized that the piecewise-constant approximation adopted in this work is not directly imposed on the original continuous multimodal signals, but rather on the event-level input representations obtained after SRM encoding. Specifically, within the SRM module, the input features first drive the accumulation of the membrane potential. Once the membrane potential reaches the firing threshold, an event is generated, and the input at that time instant is encoded into an event-level representation. This process can be summarized as $v(t) \leftarrow F(v(t^-), x(t))$,with an event occurring if $v(t) \geq \theta$, where $v(t)$ denotes the membrane potential, $x(t)$ denotes the original multimodal input feature, θ is the event-triggering threshold, and $F(\cdot)$ represents the accumulation or update process of the membrane potential driven by the input.

Let u_e denote the event-level input representation generated by the SRM at event time t_e. Then, within the adjacent event interval (t_e, t_{e+1}), the event-driven input can be approximated using a zero-order hold: $u(s) \approx u_e,\ s \in (t_e, t_{e+1})$. Accordingly, the state update equation can be simplified as

$$h(t_{e+1}^-) = \Phi(\Delta t_e)h(t_e^+) + \Gamma(\Delta t_e)u_e, \tag{7}$$

where $\Phi(\Delta t) = e^{A\Delta t}$ is the state transition matrix, and $\Gamma(\Delta t_e) = \int_0^{\Delta t_e} e^{A(\Delta t_e - s)}B ds$ is the input contribution matrix.

This approximation is reasonable in the event-driven setting. The inter-event interval, Δt_e, is not fixed by a sampling rate; rather, it is set adaptively by the SRM's threshold-triggering mechanism. When input variations push the membrane potential to the firing threshold more quickly, events occur more densely, and Δt_e shortens. Thus, the piecewise-constant approximation only applies to short local event intervals, avoiding neglect of input variations within a long fixed window.

To further characterize the approximation error, suppose that the event-level input locally satisfies Lipschitz continuity within the interval (t_e, t_{e+1}), namely $|u(s) - u_e| \le L_e(s - t_e)$, where L_e denotes the local variation intensity of the event-level input within this interval. Then, the error in the input contribution term induced by the piecewise-constant approximation satisfies $\left\| \int_{t_e}^{t_{e+1}} e^{A(t_{e+1}-s)} B\big(u(s) - u_e\big) ds \right\| \le \frac{1}{2} M_e \|B\| L_e \Delta t_e^2$, where $M_e = \max_{\tau \in [0, \Delta t_e]} \|e^{A\tau}\|$. This error bound indicates that, under the condition that the event-level input is locally smooth, the approximation error introduced by the piecewise-constant assumption is quadratically related to the adjacent inter-event interval Δt_e. Consequently, when rapidly varying inputs trigger denser events, the resulting shorter Δt_e helps reduce the error caused by the piecewise-constant approximation.

It should be noted that the above analysis does not imply that all high-frequency components can be retained without loss. For subtle changes that do not cause the membrane potential to reach the firing threshold, or for very fast changes beyond current event-triggering limits, the piecewise-constant approximation may still fail to capture detailed local information. Therefore, this work treats the approximation as an effective but limited means of modelling event-driven dynamics, rather than as a perfect reconstruction of the original continuous signal.

Event-Triggered Jump Updates and Trace Variables

When a new event arrives at time t_{e+1}, the input $u(t)$ undergoes a discontinuous change. We define the "jump increment" to represent the input change caused by the event: $\Delta u_{e+1} = u(t_{e+1}^+) - u(t_{e+1}^-)$. Correspondingly, the state update follows the standard jump update:

$$h(t_{e+1}^+) = h(t_{e+1}^-) + B\Delta u_{e+1} \tag{8}$$

Here, Δu_{e+1} is composed of the SRM event input increment (from multimodal spikes) and the alignment feedback increment (from subsequent energy/competition modules). Thus, "alignment constraints" are not applied post-hoc but instead directly influence the state update at event times, achieving alignment guidance at the dynamical level.

To avoid numerical integration at all time steps, we introduce trace variables to cache the convolution term in Eq. (7). Specifically, we treat: $z_{e+1} = \Gamma(\Delta t_e)u_e$ as the "trace" of the input contribution in the interval, and update it only once per event interval. Since $\Phi(\Delta t)$ and $\Gamma(\Delta t)$ depend only on the time interval Δt, typical values of Δt can be cached or computed quickly. When A has a sparse or diagonalizable structure, the matrix exponential $\Phi(\Delta t)$ can be efficiently computed, making the inference cost approximately linear in the number of events E.

This event-driven analytic propagation both improves numerical stability by avoiding step-size sensitivity. It enablies direct, differentiable injection of alignment feedback into continuous dynamics, yielding a unified latent state for hierarchical contrastive objectives with complexity scaling in the number of events.

Although SynNeura also integrates event representations with continuous-time dynamics, its core idea differs fundamentally from a Neural CDE with an SNN encoder. In Neural CDE-based methods, spike events serve as encoded observation signals for evolving latent trajectories via numerical integration. SynNeura's key distinction is that events are treated both as representational information and as fundamental computational units, directly determining when and how state propagation and updates occur.

Specifically, the proposed framework adopts an event-driven analytical propagation mechanism: the liquid state is propagated in closed form between adjacent events and is updated only upon the arrival of a new event. Therefore, event timestamps explicitly govern the computational scheduling and state transition process. This design shifts the computational complexity from the length of the temporal discretization to the number of events. As a result, SynNeura is particularly suitable for weakly supervised multimodal alignment scenarios characterized by sparse events, asynchronous modalities, and long temporal spans.

Advantages and limitations of the spike-based representation

Although an individual emitted spike is binary, the temporal representation in SynNeura is not determined by an isolated binary state. Instead, it is jointly characterized by spike timing, continuous-time membrane-potential dynamics, historical firing states, and event-driven latent-state evolution. This formulation is particularly suitable for asynchronous multimodal streams with heterogeneous sampling frequencies and irregular event intervals, as the model state can evolve continuously between adjacent events and be updated upon event arrival without imposing a uniform sampling grid. The modality-specific learnable inhibitory decay rate further improves the adaptability of neuronal dynamics to different temporal characteristics across modalities. Nevertheless, binary spike generation may introduce quantization-related information loss and reduce the precision of fine-grained continuous-valued representations. In addition, although a differentiable approximation is adopted during training, the inherently non-differentiable threshold-based firing mechanism still introduces additional optimization difficulty compared with conventional fully differentiable activations. Overall, the SRM-based representation is well suited to the asynchronous multimodal temporal alignment considered in this work because of its continuous-time and event-driven dynamics. At the same time, its binary spike generation may limit the representation of fine-grained continuous-valued information.

3.4. Hierarchical cross-modal alignment objective

Under weakly supervised conditions, multimodal temporal alignment often lacks explicit frame-level matching signals. To address this, in this section, we propose a hierarchical alignment objective based on the event-level input $V^{(m)}(t)$ and continuous-time liquid state $h^{(m)}(t)$ from Sections 3.2 and 3.3. This objective encourages cross-modal temporal consistency by imposing constraints at the event, trajectory, and state levels, thereby improving both training stability and cross-modal signal alignment.

In the event-driven framework, consider two modalities (denoted as A and B) with respective event time sets $\{t_i^A\}_{i=1}^{E_A}$ and $\{t_j^B\}_{j=1}^{E_B}$. By sampling liquid states at these event times, we obtain event representations $e_i^A = \phi(h^A(t_i^A))$ and $e_j^B = \phi(h^B(t_j^B))$, where $\phi(\cdot)$ is a projection head used to generate the latent space representations. Based on the principle of "temporal proximity + representation similarity," we propose a strategy to construct cross-modal candidate event pairs. Specifically, an event pair (i, j) belongs to the candidate set P if the event times in modalities A and B are sufficiently close and their representations exhibit high similarity:

$$P = \{(i, j) \mid |t_i^A - t_j^B| < \Delta t,\ \cos(e_i^A, e_j^B) > \tau_c\} \tag{9}$$

Here, Δt is the temporal neighborhood threshold, and τ_c is the representation similarity threshold. This strategy allows for the automatic construction of high-confidence positive sample pairs from sparse events, even in the absence of precise alignment labels.

For event-level alignment, we adopt the cross-modal InfoNCE loss. Specifically, the event i in modality A is considered as an anchor, with its corresponding positive sample event j in modality B determined from the candidate event pair set P. Negative samples are drawn from other events in modality B within the same batch. The spike-level alignment loss is defined as:

$$\mathcal{L}_{\text{spike}} = -\frac{1}{|P|} \sum_{(i,j)\in P} \log \left(\frac{\exp(\text{sim}(e_i^A, e_j^B)/\tau)}{\sum_{k\in N_B(i)} \exp(\text{sim}(e_i^A, e_k^B)/\tau)} \right) \tag{10}$$

Where $\text{sim}(\cdot,\cdot)$ represents the cosine similarity, τ is the temperature coefficient, and $N_B(i)$ denotes the candidate set in modality B associated with event i, including both positive and negative samples. This loss function encourages the proximity of positive sample pairs and the separation of negative samples, thereby establishing accurate cross-modal temporal correspondences.

However, relying solely on local event alignment may neglect global dynamic patterns. To address this, we introduce a trajectory-level consistency constraint. Within a time window $[t_s, t_e]$, we impose a dynamic consistency constraint on the liquid state trajectories in both modalities A and B. Specifically, we construct discrete trajectory sequences H^A and H^B at event times or a small number of sampling points and define the trajectory-level consistency loss using Soft-DTW: $\mathcal{L}_{\text{traj}} = \text{SoftDTW}_\gamma(H^A, H^B)$.

Where γ is the smoothing coefficient. This loss function constrains the relative shape and dynamic patterns of the two trajectories at a global level, avoiding alignment based solely on local pulse matching and accounting for the overall temporal structure.

Furthermore, to ensure smooth state evolution and suppress numerical instability, we introduce state-level dynamic regularization. This loss function smooths state transitions by controlling the magnitude of state derivatives, defined as: $\mathcal{L}_{\text{state}} = \frac{1}{T_w}\int_{t_s}^{t_e} \|\dot{h}(t)\|_2^2\, dt$.

Where $T_w = t_e - t_s$ is the time window length, and $\dot{h}(t)$ represents the derivative of the state. In the event-driven implementation, the derivative $\dot{h}(t)$ can be explicitly derived from the continuous-time dynamics model, allowing the regularization term to be efficiently computed using closed-form or piecewise approximations, avoiding numerical integration over dense time grids. This regularization term helps suppress high-frequency oscillations and state explosions, thereby improving training stability and robustness of cross-modal alignment.

Finally, by combining spike-level alignment, trajectory-level consistency, and state-level smoothness constraints, we construct the overall alignment objective, which is the weighted sum of the three loss terms:

$$\mathcal{L}_{\text{align}} = \mathcal{L}_{\text{spike}} + \lambda_1 \mathcal{L}_{\text{traj}} + \lambda_2 \mathcal{L}_{\text{state}} \tag{11}$$

Where λ_1 and λ_2 are the weighting coefficients for the respective losses. This objective enables gradient updates at event times via event-driven backpropagation, significantly reducing the computational overhead of training on long sequences.

Through this hierarchical alignment objective, we effectively promote the alignment of multimodal time series in the absence of precise alignment labels. The combined losses at the event, trajectory, and state levels enhance model stability and robustness while reducing computational demands during training, particularly for long sequences and sparse events.

These three alignment objectives address complementary challenges in multimodal temporal alignment. They do not simply combine existing loss functions. The spike-level InfoNCE objective works within local event neighborhoods to learn fine-grained semantic correspondences between cross-modal events. However, focusing solely on local event matching can lead to temporal drift and structural inconsistency. To address this, the trajectory-level Soft-DTW objective constrains the overall shape and trend of state trajectories from different modalities over longer temporal windows. This helps maintain consistent long-range temporal dynamics. Yet, trajectory matching alone does not regularize continuous-time state evolution. Therefore, we introduce state-level dynamical regularization to suppress high-frequency oscillations in liquid state evolution. This step improves numerical robustness during long-sequence propagation. In summary, these three objectives apply complementary constraints from the perspectives of local event matching, global trajectory consistency, and continuous dynamical stability. Together, they form a hierarchical alignment mechanism for weakly supervised multimodal temporal alignment.

3.5. *Event-driven learning and inference of temporal alignment paths*

In the event-driven framework presented in previous sections, the event streams output by the SRM and the liquid-state representations $h^{(m)}(t)$ provide continuous-time representations for cross-modal alignment. In Section 3.4, a hierarchical alignment objective was introduced to learn event-level similarity and trajectory-level consistency. This section focuses on the construction and inference of the alignment path: we represent cross-modal alignment as a matching path that satisfies temporal monotonicity and propose a differentiable method for learning and inferring this path during both training and inference stages.

For two modalities A and B, let their event time sets be denoted by $\{t_i^A\}_{i=1}^{E_A}$ and $\{t_j^B\}_{j=1}^{E_B}$, respectively. We construct a cross-modal event alignment graph $G = (V, E)$, where the node set V is the union of events from both modalities. The edge set E consists of candidate event pairs that satisfy both temporal proximity and representation similarity:

$$E = \left\{ (i,j) \;\middle|\; |t_i^A - t_j^B| \le \Delta t,\ \text{sim}(e_i^A, e_j^B) \ge \tau_c \right\} \tag{12}$$

Each candidate edge (i,j) indicates a potential correspondence between event i in modality A and event j in modality B.

To learn the alignment path end-to-end under weak supervision, we use trajectory-level Soft-DTW as the differentiable matching mechanism. Specifically, within the window $[t_s, t_e]$, the state trajectories of both modalities are discretized into sequences H^A and H^B, and Soft-DTW provides a smoothed alignment cost and a soft alignment matrix $P \in \mathbb{R}^{L_A \times L_B}$, where P_{pq} represents the soft weight of the match between sample points (p,q). This soft path is used both to compute the trajectory-level loss $\mathcal{L}_{\text{traj}}$ and as a prior for candidate event matches, guiding the exploration and updating of P, thus forming a consistent learning process of “local event matching — global trajectory path.”

In the inference stage, we use the soft alignment path π obtained by the event-driven Soft-DTW algorithm, and optimize it through dynamic programming on the cost matrix $C = [c_{ij}]$, where $c_{ij} = \text{sim}(e_i^A, e_j^B)$, and impose a temporal proximity constraint on adjacent event timestamps: $|t_i^A - t_j^B| \le \Delta t$.

Regarding the event-driven training implementation, since both model state updates and representation sampling are centered around event times (as seen in Sections 3.2 and 3.3), the necessary states and loss terms are computed only at event times and their adjacent intervals, avoiding the need to unfold over dense time grids. It is important to emphasize that gradients are still backpropagated end-to-end through differentiable dynamical propagation and alignment losses. The “event-driven” nature manifests as follows: state evolution is performed using piecewise-analytic propagation, and losses are evaluated at event times or sampling points, thereby significantly reducing computational and storage overhead during long-sequence training.

This methodology enables efficient end-to-end learning of the alignment path, leveraging both event-level matching and trajectory-level consistency. Enforcing temporal monotonicity and causality constraints ensures that the alignment is physically plausible while optimizing the cross-modal alignment process through differentiable mechanisms.

3.6. *Complexity and stability analysis*

Event-driven inference transforms the state-update mechanism of continuous-time dynamical systems from conventional fixed-step numerical integration to event-triggered analytical propagation, so that computational complexity scales primarily with the number of events rather than the size of the temporal grid.

The SRM's membrane-potential triggering mechanism determines the event density in SynNeura. Variations in input features drive the membrane potential to accumulate over time; once the potential reaches the firing threshold, an event is generated, and the hidden state is updated accordingly. Thus, unlike fixed-time-step methods, SynNeura does not update its states uniformly according to a predefined sampling frequency. Instead, state updates are adaptively triggered by input-induced membrane-potential responses.

For visual or speech patterns with rapid variations that continuously drive the membrane potential across the threshold, events are generated more frequently, leading to a higher state-update frequency. In contrast, in stationary or redundant regions, the membrane potential is less likely to reach the threshold, resulting in fewer events and thus reducing unnecessary computation. Consequently, event density can, to some extent, reflect the local dynamic intensity of the input signal, allowing the model to allocate higher temporal resolution to dynamically salient segments while suppressing redundant updates in stationary regions. This mechanism helps mitigate the approximation errors introduced by piecewise-constant representations in rapidly changing regions.

Nevertheless, this mechanism does not guarantee complete preservation of all high-frequency variations. Subtle changes below the firing threshold, or high-frequency components that oscillate rapidly over very short time spans but fail to elicit effective event responses, may still be lost in the event-level representation. We therefore further discuss this limitation in the manuscript and identify adaptive event thresholds, local linear approximations, and higher-order interpolation as promising directions for enhancing the model's ability to capture rapid local dynamics.

Let N_e denote the total number of events in $[0, T]$ and d the state dimensionality. In each inter-event interval, one propagation operator (Eq. (7)) and one jump update (Eq. (8)) are applied, resulting in an overall complexity of $O\left(N_e \cdot C_{\exp}(d)\right)$, where $C_{\exp}(d)$ is the cost of computing and applying the propagation operator, typically bounded by $O(d^2)$ and often reduced under structural sparsity.

By contrast, step-based numerical integration incurs $O(N_t \cdot C(d))$, where N_t is the number of time steps and generally $N_t \gg N_e$ for long, sparse sequences, underscoring the computational advantage of the event-driven scheme.

To analyze numerical stability, we consider the linearized form of the continuous-time dynamics. Let $x(t)$ denote the abstract state governed by the linearized system, whose dynamics are given by $\dot{x}(t) = Ax(t)$, where the state matrix A is defined as $A = \frac{1}{\tau}(U - \alpha I)$. Here, τ is the time constant, U denotes the internal coupling weight matrix, and $-\alpha I$ is an explicit dissipative leakage term that contributes to the stability of the system. For this homogeneous linear system, if the largest real part of the eigenvalues of A satisfies $\max_i \mathrm{Re}\left(\lambda_i(A)\right) < 0$, then the system is exponentially stable.

In this work, the internal coupling matrix U is initialized with an explicit spectral-radius constraint to satisfy the above stability condition. Specifically, the state matrix of the liquid layer is defined as $A = \frac{1}{\tau}(U - \alpha I)$, where $\tau > 0$ is the time constant, $\alpha > 0$ is the leakage decay coefficient, and U denotes the internal recurrent coupling matrix of the liquid layer. For a continuous-time linear system, the condition for exponential stability is $\max_i \mathrm{Re}\left(\lambda_i(A)\right) < 0$.

The internal coupling matrix is drawn from the Xavier uniform distribution: $U_0 \sim \mathrm{XavierUniform}$, *and then apply spectral normalization to it:* $U = r_{\mathrm{target}} \frac{U_0}{\rho(U_0)+\epsilon}$, where $\rho(U_0)$ denotes the spectral radius of the initial matrix U_0, ϵ is a small positive constant introduced to avoid division by zero, and r_{target} is the prescribed target spectral radius. To satisfy the stability constraint, we set $0 < r_{\mathrm{target}} < \alpha$. After spectral-radius normalization, we have $\rho(U) = r_{\mathrm{target}} \frac{\rho(U_0)}{\rho(U_0)+\epsilon} \leq r_{\mathrm{target}}$. Moreover, since any eigenvalue satisfies $|\lambda_i(U)| \leq \rho(U)$, it follows that $\mathrm{Re}\left(\lambda_i(U)\right) \leq \left|\lambda_i(U)\right| \leq \rho(U) \leq r_{\mathrm{target}} < \alpha$. Furthermore, because $\lambda_i(A) = \frac{1}{\tau}\left(\lambda_i(U) - \alpha\right), \qquad \tau > 0$, we obtain $\mathrm{Re}\left(\lambda_i(A)\right) = \frac{1}{\tau}\left(\mathrm{Re}\left(\lambda_i(U)\right)\alpha\right) < 0$. Consequently, $\max_i \mathrm{Re}(\lambda_i(A)) < 0$, indicating that the linearized liquid dynamics associated with A satisfies the exponential stability condition. Therefore, by combining Xavier uniform initialization with spectral-radius normalization and setting $0 < r_{\mathrm{target}} < \alpha$, the proposed initialization guarantees the exponential stability of the corresponding linearized liquid dynamics. During training, we further maintain spectral-radius normalization to constrain U such that $\rho(U) \leq r_{\mathrm{target}} < \alpha$, thereby preventing parameter updates from violating the exponential stability condition.

Under this condition, the matrix-exponential propagation operator satisfies $\left|e^{A\Delta t}\right| \leq Ke^{-\kappa\Delta t}$, for some positive constants $K \geq 1$ and $\kappa > 0$. This indicates that perturbations decay exponentially over time. When A is symmetric negative definite, one can set $K = 1$.

In the actual event-driven model, the liquid state $h(t)$ evolves via inter-event propagation and jump updates as described in Section 3.3. While the linearized analysis omits input-driven and nonlinear feedback terms, it offers insight into the boundedness of the analytic propagation operator. A state-level regularizer $\mathcal{L}_{\mathrm{state}}$ further suppresses high-frequency fluctuations along the trajectory, enhancing numerical stability during training.

Causality is naturally preserved: $h(t)$ depends only on past events. During backpropagation, gradients flow along the event sequence without unfolding a dense time grid, thereby reducing memory usage and improving scalability for long sequences.

This complexity analysis further highlights the distinction between SynNeura and fixed-step continuous-time models. Conventional Neural ODE and Neural CDE methods typically require numerical integration, meaning the model's behavior is approximated step by step over dense temporal grids—a set of closely spaced time points that cover the relevant interval. Their computational complexity increases with the number of temporal steps, N_t. In contrast, SynNeura's state propagation is event-driven. In sparse-event scenarios, propagation and jump updates are performed only over neighborhoods linked to N_e events. Typically, in long-sequence conditions, $N_e \ll N_t$. This lets SynNeura reduce redundant computation while still supporting continuous-time modeling. This advantage does not mean our method is a totally separate paradigm from Neural CDEs, SNNs, or liquid neural dynamics. Instead, SynNeura structurally integrates and extends these ideas. It does this through an event-driven analytical propagation mechanism for weakly supervised multimodal temporal alignment.

4. Experimental

To comprehensively evaluate SynNeura's effectiveness and generalization, we conduct extensive experiments across multiple multimodal temporal alignment benchmarks. We aim to verify whether the proposed event-driven formulation achieves higher alignment accuracy, stronger temporal consistency, improved interpretability, and better computational efficiency than representative frame-based or continuous-time baselines.

We further evaluate SynNeura under weak supervision to assess its ability to discover cross-modal temporal correspondences without dense frame-level annotations. In the following subsections, we first introduce the experimental setup and then report quantitative comparisons, qualitative analyses, and evaluations on three challenging multimodal datasets.

Accordingly, our experiments are organized around four research questions:

(RQ1) Can SynNeura accurately align fine-grained temporal positions across modalities?

(RQ2) Does SynNeura learn a correct and reliable cross-modal alignment structure?

(RQ3) Are the predicted alignment sequences temporally coherent and stable?

(RQ4) Can SynNeura accurately detect and align semantic events at the segment level?

Together, RQ1–RQ4 provide a structured evaluation framework to analyze SynNeura from the perspectives of alignment accuracy, structural consistency, temporal stability, and event-level reliability.

4.1. Experimental setup

Datasets

To evaluate the effectiveness of the proposed event-driven liquid neural dynamics model for multimodal temporal alignment, we conduct systematic experiments on three representative benchmarks:

(1) Audio-Visual Event(AVE), an audio–visual event localization dataset consisting of real-world videos across multiple categories with corresponding audio event annotations.

(2) Lip Reading Sentences2(LRS2), a large-scale lip-reading dataset comprising aligned speech and lip video clips extracted from BBC broadcasts, widely used for studying audio–visual synchronization and cross-modal alignment.

(3) YouCook2, a large-scale cooking-video understanding dataset providing step-wise temporal segment annotations with textual descriptions, commonly used for video segmentation and video captioning.

These datasets cover a broad spectrum of alignment scenarios, ranging from short-term audio–visual synchronization to long-horizon semantic activity alignment (e.g., audio–video synchronization, speech–lip motion alignment, and speech–text alignment).

Weakly Supervised Setting

To avoid ambiguity about the level of supervision, we explicitly define the weakly supervised setting in this work as follows: during training, no precise frame- or event-level cross-modal temporal alignment annotations are used. Instead, the model learns cross-modal temporal correspondences only from the coarse-grained semantic labels or segment-level descriptions already available in the datasets. Specifically, on the AVE dataset, we use video-level event category labels and segment-level event presence information, without precise audio-visual event boundary annotations. On the LRS2 dataset, we use sentence-level speech transcriptions or paired video-speech clips, without using frame-level phoneme labels, lip-motion annotations, or precise audio-video synchronization labels. On the YouCook2 dataset, we use video-segment-level textual descriptions and step-level procedural annotations, without using frame-level action boundaries or exact step-to-frame correspondence annotations. All baseline methods and SynNeura are trained and evaluated under the same weakly supervised setting to ensure a fair comparison.

Parameter Settings

In the experiments, dataset-specific hyperparameters are configured. The temporal proximity threshold Δt is set to 0.5 s, 0.4 s, and 1.0 s for AVE, LRS2, and YouCook2, respectively, reflecting differences in event-time distributions. Other hyperparameters remain consistent across datasets: the similarity threshold $\tau_c = 0.6$, InfoNCE temperature $\tau = 0.07$, and Soft-DTW smoothing $\gamma = 0.1$. The trajectory loss weight $\lambda_1 = 0.5$, state regularization $\lambda_2 = 0.1$, feedback strength $\beta = 0.2$, and feedback decay time constant $T_g = 1.0$.

For temporal modelling, the model adopts a unified set of temporal control parameters, referred to as Sequential Control (SC), where γ ranges from 40 to 250ms and τ ranges from 0.07 to 0.1s. The Delay-Time Control (DLC) range is set to 0.2–2.0s, with an allowable temporal lag range of ±0.12–0.6s. Cross-modal alignment is measured using a non-permutation simultaneous coefficient similarity metric. The input modalities are first transformed into sparse event sequences by the SRM front-end, then processed by the liquid-state module using piecewise-exact solutions to achieve stable continuous-time modelling. These settings ensure consistency in experimental configurations across datasets and improve the reproducibility of results.

Evaluation Metrics

We evaluate multimodal temporal alignment across three datasets from complementary aspects: event-boundary accuracy and cross-modal temporal consistency on AVE, cross-modal dynamic correlation on LRS2, and event-level detection quality on YouCook2. During training, negative samples are generated by randomly perturbing event timestamps to improve contrastive robustness. Experiments are conducted across multiple modality pairs (audio–video, video–text, and audio–text) to assess generalization under asynchronous input conditions.

We employ the following five evaluation metrics:

(1) Mean Alignment Error(MAE). MAE measures the accuracy of predicted semantic event boundaries by the mean absolute error between predicted boundaries and ground-truth temporal annotations.

(2) Cross-modal Alignment Score(CAS). CAS quantifies the degree of temporal alignment between auditory and visual modalities. For the i-th sample, let $s_i(t) \in [0,1]$ denote the model-predicted alignment intensity at time step t(e.g., the output of the CAS head), and let $y_i(t) \in \{0,1\}$ be the binary frame-level event label. We define the frame-wise alignment error as $|s_i(t) - y_i(t)|$, and the corresponding frame-wise consistency as $1 - |s_i(t) - y_i(t)|$. The CAS for the i-th sample is defined as: $CAS_i = \frac{1}{T_i}\sum_{t=1}^{T_i}(1 - |s_i(t) - y_i(t)|)$.

CAS ranges in [0,1]; higher values indicate stronger agreement with the ground-truth event distribution and better temporal alignment between auditory events and visual semantics.

(3) Temporal Correlation(TC). On LRS2, we evaluate audio–visual speech alignment by measuring cross-modal dynamic consistency. Let the aligned audio feature sequence be $\{a_{i,t}\}_{t=1}^{T_i}$ and the aligned visual (lip) feature sequence be $\{v_{i,t}\}_{t=1}^{T_i}$. We define the temporal correlation of the i-th sample by the average cosine similarity: and compute the overall temporal correlation as: $TC_i = \frac{1}{T_i}\sum_{t=1}^{T_i}\frac{\langle a_{i,t}, v_{i,t}\rangle}{\|a_{i,t}\|\cdot\|v_{i,t}\|}$. A higher TC indicates stronger dynamic consistency between aligned modalities, reflecting better temporal alignment and joint modeling.

(4) Event Precision and Recall(EP/ER). EP and ER evaluate the precision and recall of event detection and temporal alignment performance.

(5) Unified Evaluation Score(UES). To compare models across datasets and heterogeneous metrics, we normalize all metrics to the range [0, 1]. The temporal error MAE is converted into a score: $S_{\text{MAE}} = \max\left(0,\ 1 - \frac{\text{MAE}}{M_{\max}}\right)$.

Other metrics (CAS, TC, EP, ER) already lie in [0,1] and are used directly. The overall unified score is computed as a weighted average: $\text{UES} = \sum_m w_m S_m$, $\sum_m w_m = 1$, where we use equal weights by default.

To systematically evaluate SynNeura across capability dimensions, we design complementary experiments covering alignment accuracy, structural reliability, temporal coherence, and robustness—quantitative benchmarks for fine-grained alignment and key segment localization (RQ1, RQ4). Qualitative visualizations examine the correctness of learned alignment structures and the coherence and stability of inferred trajectories (RQ2, RQ3), while also illustrating event-level alignment behavior (RQ4). Ablation and sensitivity studies assess robustness under architectural variations and perturbations, and quantify the contribution of each component (RQ2, RQ3, partially RQ1).

4.2. Benchmarking and quantitative evaluation

To ensure that comparisons among different methods in the multimodal temporal alignment task are fair, reproducible, and sufficiently interpretable, we implement several representative temporal dynamics modeling methods within a unified task setting, with a common input source, output format, training protocol, and evaluation metric. These methods cover multiple modeling paradigms, including discrete-time Transformers, continuous-time RNNs, and spiking neural networks. Specifically, we select ClipBERT, DTW-Transformer, ODE-RNN, LTC, SRM-SNN, and Spiking-Liquid as baseline methods, and incorporate them into a unified comparison pipeline through necessary model-interface adaptations.

It should be clarified that the “unified framework” in this work does not mean that all baseline methods are forcibly converted into the same event-driven model or continuous-time model, nor does it imply that

Table 1
Capability coverage of baseline models for weakly supervised multimodal temporal alignment.

Method	Async/Delay	Weak Sup.	Continuous-time	Event-driven	Long-seq Scalability	Interpretability
ClipBERT	Δ	Δ	✗	✗	Δ	Δ
DTW-Transformer	Δ	Δ	Δ	✗	✗	Δ
ODE-RNN	✓	Δ	✓	✗	Δ	Δ
LTC	✓	Δ	✓	✗	Δ	Δ
SRM-SNN	Δ	Δ	Δ	✓	Δ	✓
Spiking-Liquid	Δ	Δ	✓	✓	Δ	✓
SynNeura(Ours)	✓	✓	✓	✓	✓	✓

Note: (✓: explicitly supported; Δ: partially supported / implementation-dependent; ✗: not supported). “Async/Delay” denotes explicit cross-modal delay/asynchrony modeling; “Weak Sup.” denotes compatibility with video-/segment-level supervision without dense alignment labels; “Continuous-time” denotes ODE/CDE/LTC-style dynamics (vs. fixed-grid discretization); “Event-driven” denotes event/spike-triggered computation; “Long-seq” denotes stability/efficiency under long, low-sampling, event-sparse sequences; “Interpretability” denotes event-/trajectory-level explanations (e.g., triggers and state evolution).

Table 2
Quantitative comparison across multimodal alignment benchmarks.

Model	AVE		LRS2	YouCook2	mean performance↑
	MAE↓	CAS↑	TC↑	EP↑ / ER↑	
ClipBERT	0.421±0.018	0.612±0.025	0.537±0.030	0.782 / 0.701	0.570
DTW-Transformer	0.398±0.016	0.635±0.022	0.563±0.027	0.803 / 0.720	0.600
ODE-RNN	0.364±0.013	0.654±0.021	0.574±0.025	0.811 / 0.738	0.620
LTCA	0.352±0.014	0.667±0.018	0.589±0.024	0.820 / 0.743	0.640
SRM-SNN	0.331±0.012	0.682±0.020	0.603±0.023	0.836 / 0.759	0.660
Spiking-Liquid	0.308±0.011	0.706±0.018	0.621±0.020	0.850 / 0.772	0.680
SynNeura(Ours)	0.271±0.009	0.713±0.015	0.658±0.017	0.889 / 0.824	0.730

Table 3
Partial runtime efficiency comparison under the same hardware environment, batch size, and input length.

Method	Inference Time (ms) ↓	FLOPs (G) ↓	GPU Memory (GB) ↓	Parameters (M) ↓
ClipBERT	42.6	58.4	4.8	112.3
DTW-Transformer	36.8	42.7	3.9	86.5
ODE-RNN	31.5	34.2	3.1	54.8
LTC	28.9	29.6	2.8	47.2
SRM-SNN	24.7	22.5	2.4	39.6
Spiking-Liquid	22.3	19.8	2.2	36.9
SynNeura	20.1	17.4	2.0	34.5

all methods adopt the same preprocessing pipeline or internal representation. Since ClipBERT, DTW-Transformer, ODE-RNN, LTC, SRM-SNN, and Spiking-Liquid differ in their input formats, modeling assumptions, and temporal modeling mechanisms, we do not enforce a single preprocessing module across all methods. Instead, we ensure that all methods use the same data splits, raw samples, basic modality feature sources, supervision signals, task output head, loss function, training settings, and evaluation metrics. In this setting, each method undergoes only the interface adaptation necessary for its input format.

Specifically, for the audio, video, and text modalities, we first extract basic modality-specific features from the same data splits and raw samples. Then, according to the input format required by each method, we construct the corresponding input representations from the same basic modality features and annotation information, such as sequential features, timestamped sequences, continuous-time inputs, or spike-encoded inputs. Here, the term “same basic input source” means that all methods are built upon the same raw samples and basic modality features before entering their respective temporal modeling modules. It does not imply that the hidden states, intermediate features, or final representations learned internally by different methods are identical.

For different types of baseline methods, we preserve their original modeling assumptions as much as possible. ClipBERT retains its video-language input organization and representation-modeling structure, with basic visual and language features mapped to its video-language modeling interface. DTW-Transformer constructs sequence-alignment inputs from the same basic sequential features and preserves its dynamic time warping and Transformer-based sequence modeling mechanisms. ODE-RNN and LTC introduce the corresponding timestamps or time intervals on top of the same basic input sequences to satisfy the requirements of continuous-time state modeling. SRM-SNN performs spike encoding based on the same basic feature sources to meet the requirements of spiking neural networks for discrete spike inputs. Spiking-Liquid constructs spike inputs or liquid-state inputs from the same basic feature sources and preserves its spiking liquid-state dynamics modelling mechanism.

It should be particularly emphasized that the SRM front-end, liquid-state propagation, and trace-variable mechanisms specific to SynNeura are not equally added to all baseline methods. For non-spiking or non-event-driven methods such as ClipBERT, DTW-Transformer, ODE-RNN, and LTC, we do not introduce an additional SRM front-end, as doing so would alter their original modeling assumptions and may introduce new unfair factors. For spiking baselines such as SRM-SNN and Spiking-Liquid, we retain only the spike-encoding or spiking-state propagation mechanisms required by their respective architectures, without incorporating the complete SRM–liquid-state propagation–trace-variable modeling pipeline of SynNeura.

Therefore, the experimental comparison in this work is not conducted under completely identical preprocessing pipelines or internal

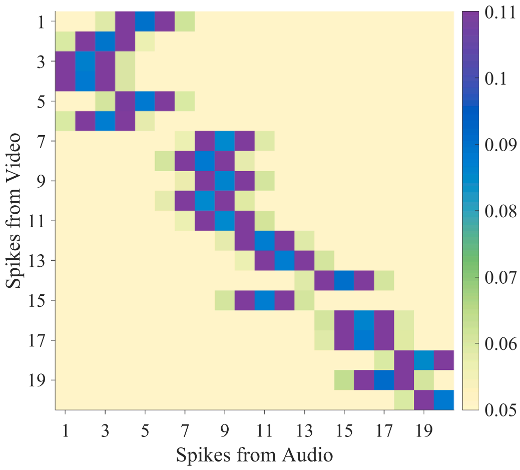


Fig. 2. Spike-to-spike optimal transport plan.

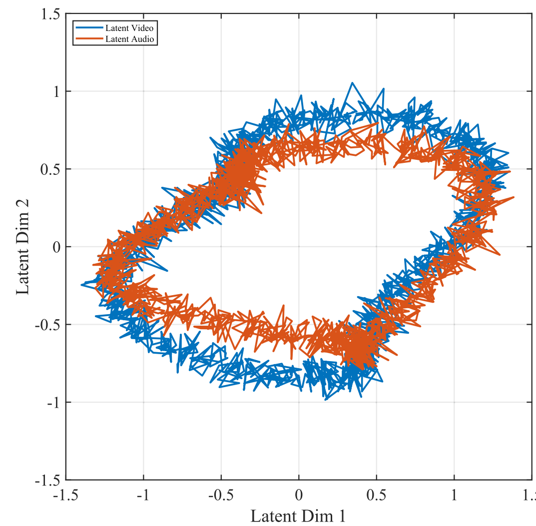


Fig. 3. Latent trajectories for STC alignment.

representations. Instead, it evaluates the performance of different temporal dynamics modeling mechanisms for multimodal temporal alignment using the same data sources, basic modality features, supervision signals, task output setting, and evaluation metrics. This setup ensures consistent comparison conditions while avoiding imposing method-specific modules from one approach onto other baselines, thereby reducing potential bias from excessive adaptation.

To guarantee strict comparability, we employ identical task heads and loss functions across all methods:

- AVE: event timestamp regression and alignment evaluation (MAE/CAS).
- LRS2: audio–visual speech alignment evaluation via temporal correlation and consistency (TC).
- YouCook2: segment-level event detection evaluation (EP/ER). We additionally report UES as a supplementary metric to provide a unified comparison across heterogeneous criteria.

Importantly, the only difference among baselines lies in their internal mechanisms for modeling temporal dynamics, whereas input representations, task definitions, output heads, and evaluation protocols are kept identical. Since prior work has not evaluated these methods on AVE, LRS2, and YouCook2 under a unified metric protocol, our standardized implementation enables controlled comparisons. It improves the fairness and interpretability of the results. Table 1 summarizes the baselines and highlights their coverage across key capability dimensions, including asynchrony and delay handling, weak-supervision compatibility, continuous-time modeling, event-triggered computation, long-sequence scalability, and interpretability.

As summarized in Table 1, the selected baselines provide broad coverage of key temporal modeling capabilities, including explicit asynchrony and latency handling, compatibility with weak supervision, continuous-time dynamics, event-driven computation, scalability to long sparse sequences, and interpretability at the event and trajectory level.

Table 2 summarizes the quantitative results of all compared methods on the AVE, LRS2, and YouCook2 multimodal temporal alignment benchmarks. SynNeura consistently outperforms both continuous-time models (ODE-RNN, LTC) and spiking approaches (SRM-SNN, Spiking-Liquid) across all metrics.

On AVE, SynNeura reduces the event-timestamp MAE from 0.308 (best baseline) to 0.291 and improves CAS to 0.713, indicating higher event-level temporal precision and stronger segment-wise cross-modal consistency. On LRS2, which emphasizes frame-level audio–lip synchronization, SynNeura achieves a TC of 0.648, outperforming ODE-RNN, LTC, and Spiking-Liquid, and demonstrating better preservation of temporally consistent cross-modal dynamics. On the long-horizon procedural dataset YouCook2, SynNeura improves EP/ER to 0.829/0.794, achieving consistent gains in both precision and recall for key-step detection, consistent with its ability to capture sparse yet semantically critical events across extended sequences.

These improvements can be attributed to SynNeura's event-driven liquid spiking dynamics. First, the model performs strong state updates only when salient cross-modal events occur, while maintaining smooth evolution during non-event intervals; this suppresses noise and preserves useful temporal memory, which is particularly beneficial for sparse-event scenarios such as AVE and YouCook2. Second, learnable liquid time constants enable adaptation across temporal scales, supporting both short-range synchronization on LRS2 and long-range semantic alignment on YouCook2. The unified evaluation score UES further confirms that SynNeura achieves the highest average performance across the three datasets, indicating robust and generalizable alignment beyond benchmark-specific tuning.

To further validate the computational characteristics of SynNeura, we conduct a partial runtime efficiency comparison under the same hardware environment, batch size, and input length. We report the wall-clock inference time, peak GPU memory consumption, number of parameters, and estimated FLOPs. For dynamic-computation models such as SynNeura, SNN-based models, ODE-RNN, and LTC, the actual computational cost can be affected by the event density, the number of time steps, or the number of continuous-time solver steps. Therefore, we explicitly mark FLOPs as estimates and compute them using the average event density of the test set and a unified input length, primarily to provide a relative reference for computational cost.

As shown in Table 3, under the same hardware environment, batch size, and input length, SynNeura demonstrates favorable runtime efficiency in terms of inference time, estimated FLOPs, GPU memory consumption, and number of parameters. Compared with dense sequence modeling methods such as ClipBERT and DTW-Transformer, SynNeura reduces the inference time from $42.6\mathrm{ms}$ and $36.8\mathrm{ms}$ to $20.1, \mathrm{ms}$, respectively. The estimated FLOPs are reduced from 58.4G and 42.7G to 17.4G, while the GPU memory consumption decreases from 4.8GB and 3.9GB to 2.0GB. Compared with continuous-time baselines such as ODE-RNN and LTC, SynNeura also exhibits lower inference overhead. This improvement primarily stems from its event-driven propagation mechanism, in which major state updates occur only when key spike events are detected, rather than requiring continuous integration over dense temporal grids or full-sequence updates.

It is worth noting that FLOPs are estimated using a unified profiling tool with the same input length and the test set's average event density,

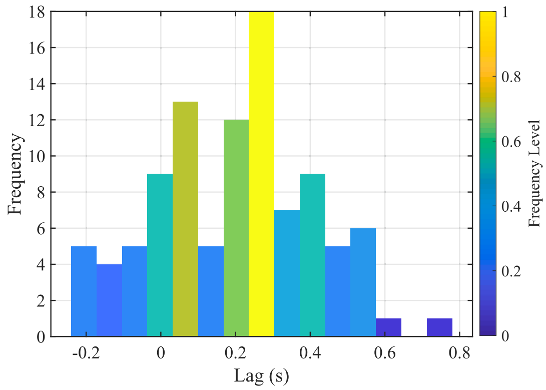


Fig. 4. Histogram of learned lags.

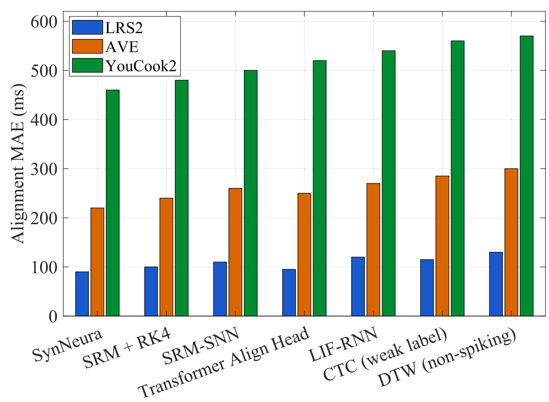


Fig. 5. Comparison of alignment accuracy(lower is better).

and are primarily intended to provide a relative comparison of computational cost across methods. For dynamic-computation models such as SynNeura, SNN-based models, ODE-RNN, and LTC, the actual FLOPs may vary with event density, the number of time steps, or the number of continuous-time solver steps. Therefore, the efficiency-related conclusions are limited to empirical observations in the current experimental setting, rather than to hardware-agnostic absolute claims. Overall, this runtime efficiency analysis provides additional empirical support for SynNeura's potential computational advantages in event-sparse and long-sequence multimodal alignment scenarios.

After establishing the main advantages under the unified alignment framework, we conduct a finer-grained analysis of the temporal dynamics in SynNeura. To isolate the contribution of the event-driven liquid mechanism, we introduce controlled variants and reference methods: (i) SRM + RK4, spike-response dynamics integrated via a fixed-step RK4 solver; (ii) SRM-SNN, a standard SRM-based SNN with fixed decay kernels; (iii) LIF-RNN, recurrent LIF dynamics representing classical SNN behavior; (iv) Transformer Align Head, a frame-based Transformer with a cross-modal alignment head; (v) DTW, non-parametric dynamic time warping; and (vi) CTC, a differentiable monotonic alignment objective without explicit dynamics. This controlled set disentangles the effects of spiking dynamics, numerical solvers, and alignment formulations, enabling clearer attribution of SynNeura's performance gains.

Fig. 2 illustrates the optimal transport (OT) plan matrix for cross-modal spike matching on AVE, where brightness indicates the transport weight. The transport mass mainly falls within 0.05–0.11. A prominent high-intensity band concentrated around the main diagonal reveals an approximately one-to-one correspondence between audio and visual events, indicating that the two modalities are largely synchronized in temporal rhythm. The continuous dark diagonal structure suggests stable cross-modal coupling. In contrast, a small number of off-diagonal blocks correspond to local temporal shifts or rhythm modulations, capturing transient temporal compression and expansion.

Several isolated strong responses away from the diagonal indicate occasional many-to-one mappings induced by noise or transient events, suggesting local desynchronization or phase drift. Overall, the OT plan exhibits a clear pattern of "global alignment with local perturbations": the dominant matching path is compact and stable, while peripheral regions remain low-weight. These observations suggest that the SynNeura–OT framework preserves global temporal consistency while flexibly adapting to fine-grained dynamics, providing robust and interpretable multimodal synchronization.

To visualize the learned latent dynamics, we project the high-dimensional hidden states $z_t \in \mathbb{R}^D$ onto a 2D subspace using PCA, and denote the axes as Latent Dim 1 and Latent Dim 2.

Fig. 3 shows the projected latent trajectories of the Video and Audio modalities, providing an intuitive view of the dynamic subspace consistency encouraged by the spatio-temporal consistency (STC) mechanism. Both trajectories form approximately ring-shaped (elliptical) patterns composed of densely connected, slightly jagged curves, reflecting discretely sampled oscillatory dynamics. The overall circular geometry suggests a cyclic latent structure rather than convergence to a static state.

The two trajectories share a similar topology but are not perfectly aligned: the red (audio) trajectory lies slightly within the blue (video) trajectory and intersects it locally. Such spatial deviations are consistent with temporal delays and phase offsets—i.e., the modalities evolve on a shared latent manifold with subtle timing differences. Overlapping regions correspond to local synchrony, whereas separated segments indicate transient misalignment, again reflecting globally coherent yet locally perturbed dynamics. The unsmoothed, jagged boundaries preserve short-timescale fluctuations, highlighting stochastic variations in the latent process. Overall, Fig. 3 provides qualitative evidence that STC maintains global dynamical coherence while remaining adaptive to fine-grained temporal discrepancies.

To comprehensively evaluate the proposed framework's capability to capture temporal dependencies and achieve cross-modal synchronization, we analyze its behavior from two complementary perspectives: the learned temporal lag distribution and comparative evaluations against multiple alignment methods. These analyses jointly characterize the model's temporal modeling ability, alignment accuracy, and adaptivity.

Fig. 4 illustrates the temporal lag distribution learned during alignment. The distribution is asymmetric yet highly concentrated, with a dominant peak in 0.1–0.3s and a mode around 0.2s, indicating that the model consistently learns a stable optimal offset (approximately 200ms) in most cases. Long tails in the negative-lag region (≈ -0.1 s) and in larger positive lags ($> 0.4s$) suggest that a subset of samples exhibits temporal drift or asymmetric alignment patterns, potentially caused by modality-dependent sampling discrepancies or transient phase shifts. Overall, the lag distribution reflects SynNeura's globally consistent yet locally adaptive behavior in dynamic temporal alignment.

Fig. 5 compares multiple alignment approaches on AVE, LRS2, and YouCook2. The x-axis denotes model type, and the y-axis reports alignment error (MAE, ms; lower is better). SynNeura achieves the lowest MAE across all tasks, outperforming SRM + RK4, SRM-SNN, LIF-RNN, Transformer Align Head, as well as classical DTW and CTC, which are less effective at capturing nonlinear temporal dependencies under weak supervision and in non-event-driven settings.

In summary, by integrating event-driven updates with continuous-time propagation, SynNeura substantially reduces alignment error and improves temporal consistency, demonstrating strong generalization and robustness across benchmarks. These results support SynNeura as an effective paradigm for cross-modal dynamic synchronization and unified temporal representation learning.

Fig. 6 compares seven models (SynNeura, SRM + RK4, SRM-SNN, LIF-RNN, Transformer Align Head, DTW, and CTC) on AVE, LRS2, and YouCook2. We report Alignment MAE (ms; lower is better) with Std

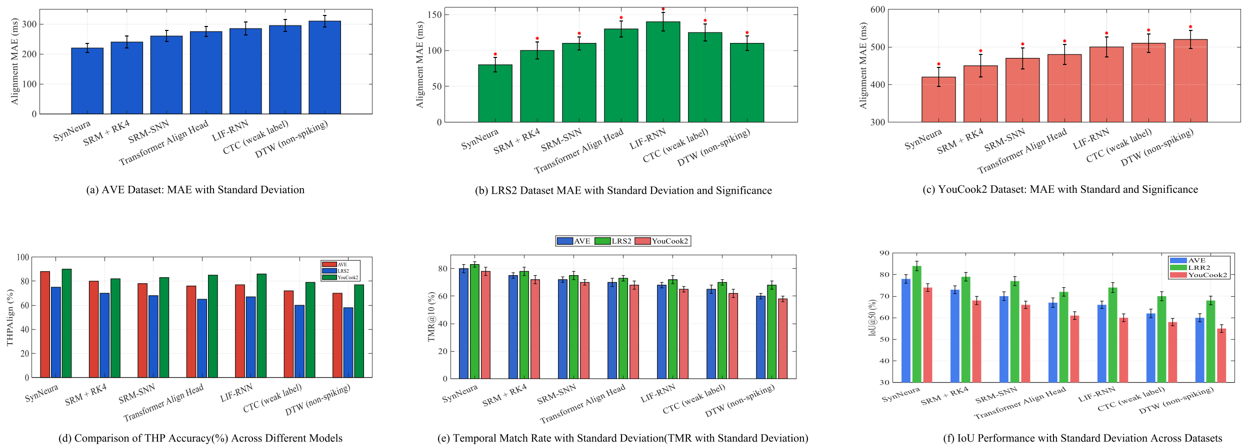


Fig. 6. Comparison of time alignment performance across different models on multimodal datasets.

(black error bars), and mark statistically significant differences from SynNeura with red asterisks ($p < 0.05$).

Across AVE (Fig. 6a), LRS2 (Fig. 6b), and YouCook2 (Fig. 6c), SynNeura achieves the lowest MAE and the smallest variance, indicating superior accuracy and stability under cross-modal asynchrony and temporal drift. In contrast, SRM + RK4 and SRM-SNN exhibit consistently higher errors and larger fluctuations, suggesting limited temporal adaptivity without liquid modulation. LIF-RNN and Transformer Align Head achieve moderate accuracy but show reduced stability on long sequences, while DTW and CTC perform worst, particularly in the more complex YouCook2 setting.

Beyond MAE, SynNeura attains the highest Temporal Match Rate (TMR) across datasets (Fig. 6d and e), reaching approximately $\geq 75\%$ on AVE and LRS2 and maintaining a clear lead on YouCook2, with the smallest Std. Moreover, SynNeura achieves the best IoU@0.5 with the lowest variance (Fig. 6f), especially under long-horizon and semantically complex alignment in YouCook2.

Overall, these results indicate that event-driven liquid-spiking dynamics enable SynNeura to deliver high-precision, low-variance temporal synchronization, with strong cross-modal consistency and cross-domain robustness.

In multimodal temporal alignment, different models can exhibit substantial performance gaps, particularly in terms of accuracy and robustness. To provide a holistic comparison, Figs. 7 and 8 analyse seven representative methods on AVE, LRS2, and YouCook2, using IoU@0.5 and the distribution of learned temporal offsets as primary indicators.

Fig. 7 summarizes the overall alignment quality of seven models (SynNeura, SRM + RK4, SRM-SNN, LIF-RNN, Transformer Align Head, DTW, and CTC) in terms of IoU@0.5. The radar plot uses three axes corresponding to the three benchmarks (AVE, LRS2, and YouCook2). The three curves correspond to audio–visual, speech–visual, and video–language alignment tasks, respectively. A larger envelope (closer to the outer boundary) indicates higher IoU@0.5 and stronger generalization. SynNeura forms the largest and most complete polygon across all axes and tasks, indicating superior alignment accuracy and cross-dataset consistency. The advantage is especially pronounced on AVE and LRS2, and remains clear on YouCook2, suggesting improved robustness under sparse events, fine-grained synchronization, and long-horizon semantic alignment. In comparison, SRM + RK4 and SRM-SNN yield smaller envelopes, while LIF-RNN and Transformer Align Head shrink further, indicating reduced capacity to capture complex cross-modal temporal dependencies. Classical DTW and CTC remain the weakest overall.

Fig. 8 further examines synchronization behavior by comparing the learned temporal offset distributions of CTC, SRM + RK4, and SynNeura based on 100 samples. The x-axis denotes temporal offset (seconds),

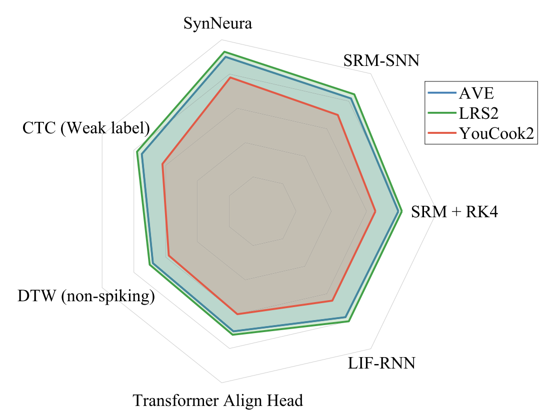


Fig. 7. The comprehensive performance comparison of different models on multimodal temporal alignment(IoU@0.5).

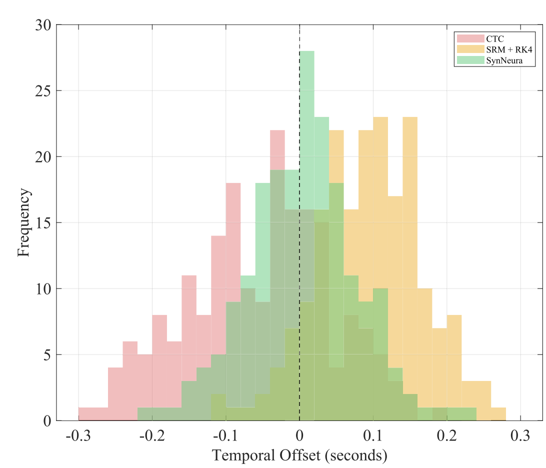
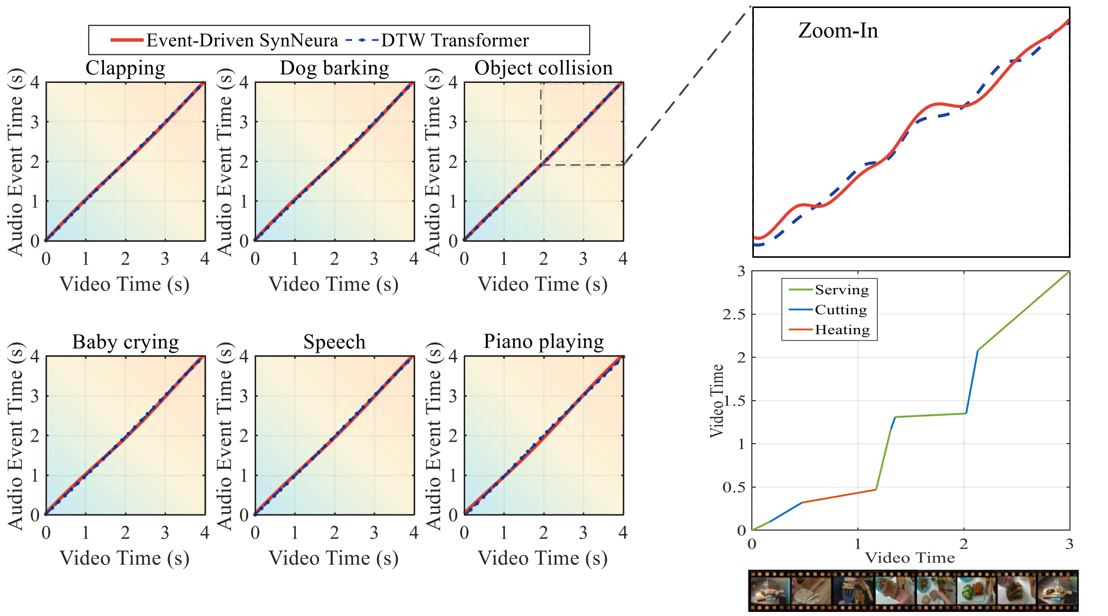


Fig. 8. Alignment delay offset distribution.

and the y-axis denotes frequency. A more concentrated, symmetric distribution centered near 0 indicates stronger synchronization and a smaller alignment error. SynNeura exhibits the sharpest and most symmetric peak around zero, indicating the smallest offset variance and the strongest temporal consistency. SRM + RK4 shows a broader distribution with a slight positive shift, consistent with mild temporal lag and



(a) Temporal alignment examples on AVE

(b) Event-aligned temporal synchronization between YouCook2 segments

Fig. 9. Overview of event-driven temporal alignment across AVE and YouCook2 datasets.

reduced adaptivity to asynchrony. CTC produces the most dispersed and asymmetric distribution (approximately −0.3s to 0.3s), reflecting temporal drift and limited temporal-structure modelling.

Overall, Figs. 7 and 8 indicate that event-driven liquid-spiking dynamics enable SynNeura to achieve robust and adaptive temporal alignment across tasks and datasets, particularly under heterogeneous temporal characteristics and weak supervision.

4.3. Qualitative analysis and alignment visualization

To further assess the temporal interpretability and event-driven modeling characteristics of SynNeura, we conduct qualitative analyses and visualizations on representative multimodal alignment scenarios. Complementary to the quantitative results, these visualizations illustrate how SynNeura forms temporally coherent alignment trajectories across modalities and achieves semantic-level synchronization.

Audio–visual event alignment on AVE. SynNeura accurately captures the synchrony between audio events (e.g., clapping, dog barking, and object impacts) and the corresponding visual frames. As shown in Fig. 9(a), the learned alignment path appears as a continuous yet sparse diagonal band, while exhibiting localized nonlinear deviations near event boundaries to compensate for modality-dependent delays. In contrast, frame-attention Transformer models often produce fragmented and drifting alignment paths under asynchronous inputs. By coupling continuous-time dynamics with event-triggered updates, SynNeura adaptively models non-uniform temporal structures, yielding stable alignments whose trajectories remain explicitly traceable along the time axis.

Video–text semantic alignment on YouCook2. The YouCook2 task requires synchronizing procedural video actions with textual descriptions. As shown in Fig. 9(b), the alignment trajectory produced by SynNeura closely coincides with textual event anchors at semantic milestones such as cutting, heating, and serving, while automatically adjusting the alignment rate in ambiguous transition regions to maintain continuity and interpretability. This visualization suggests two key implications. First, it reflects the multi-stage sequential structure of instructional videos, where semantic events occur in a progressive order rather than as isolated actions; thus, reliable alignment requires modeling global cross-stage temporal dependencies beyond local frame-level cues. Second, the significant variation in stage durations highlights the importance of robustly handling variable-length events: fixed-window designs or naive frame-level classification can cause boundary shifts for short stages (e.g., cutting and serving) and overly fragmented predictions for long stages (e.g., heating).

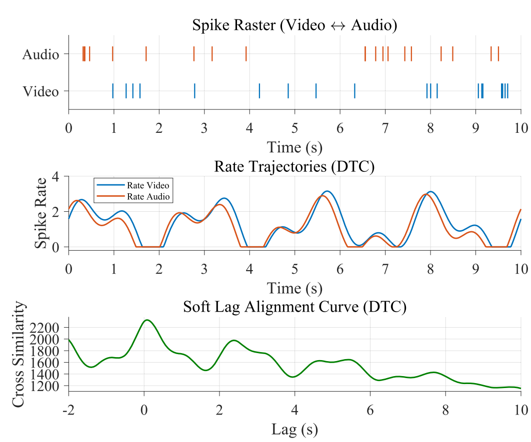


Fig. 10. Multimodal weakly-supervised temporal alignment visualization.

Overall, SynNeura produces alignment paths grounded in event times, which can be mapped to observable physical or semantic changes (e.g., peaks in audio energy, motion boundaries in video, and textual transitions). This property improves interpretability and provides semantic support for downstream tasks such as cross-modal retrieval, action segmentation, and multimodal summarization. Together, these qualitative results further support SynNeura's adaptivity and robustness for multimodal temporal alignment.

Fig. 10 provides an integrated view of temporal spiking activity and cross-modal synchronization between the Video and Audio neural modalities, including a spike raster, rate trajectories, and a soft lag alignment curve.

In the spike raster (blue: Video; red: Audio), alternating firing with partial co-activation indicates structured event-level coupling. The rate trajectories (Gaussian-smoothed) reveal largely co-occurring peaks with small phase offsets, suggesting strong rate-level correlation with subtle delay; such relationships can be quantified by Dynamic Time Correlation (DTCorr) to support lag estimation. The soft lag alignment curve plots cross-correlation versus lag: a peak near 0 indicates near-synchrony, while a rightward shift implies that Audio lags behind Video, and a leftward shift implies that Video lags behind Audio; the peak location and height reflect the optimal offset and coupling strength, respectively.

Together, these visualizations form a hierarchical analysis chain from discrete spikes to continuous dynamics and finally to quantitative lag estimation, providing interpretable evidence of cross-modal temporal dependence.

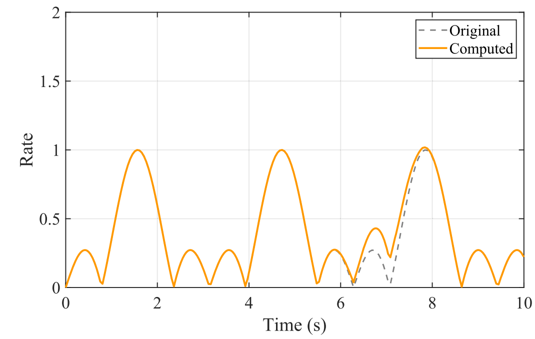


Fig. 11. Robustness to spike corruption.

4.4. Ablation and sensitivity studies

To experimentally assess the necessity of each component in SynNeura, we further design a set of systematic ablation studies by progressively removing or replacing its core modules. Specifically, we construct the following variants, each targeting a distinct aspect of the model: (i) w/o SRM, where the SRM-based event encoder is removed and frame-level features are directly used as input, to examine the necessity of event encoding; (ii) w/o event-driven propagation, where the proposed closed-form event-driven liquid propagation is replaced with fixed-step recurrent updates, to test the influence of event-driven propagation; (iii) w/o spike-level loss, where the InfoNCE constraint between local cross-modal candidate events is removed, to evaluate the impact of local event matching; (iv) w/o trajectory-level loss, where the Soft-DTW loss for enforcing global temporal trajectory consistency is removed, to assess the importance of global trajectory consistency; and (v) w/o state-level regularization, where the dynamical smoothing regularization term on liquid states is removed, to measure the effect of continuous dynamical stability. These variants are designed to evaluate the effects of event encoding, event-driven liquid propagation, local event matching, global trajectory consistency, and continuous dynamical stability on the overall model performance.

To further validate the practical contribution of each component in SynNeura, we conduct systematic ablation studies on the SRM event encoder, liquid-state propagation, trace-variable mechanism, and hierarchical alignment losses, and report the results in Table 4. Overall, the complete SynNeura achieves the best performance across all metrics: the lowest MAE (0.291), the highest CAS (0.713), the highest TC (0.658), and the best EP/ER (0.889/0.824). These results indicate that performance improvement does not arise from a single module, but from the synergistic interaction among event encoding, continuous-time liquid propagation, trace-variable memory, and hierarchical alignment objectives.

First, the SRM event encoder plays a crucial role in event-time modeling. After removing the SRM front-end, the model uses dense frame-level features directly as input, resulting in MAE increasing from 0.291 to 0.356 and TC decreasing from 0.658 to 0.594. This significant performance degradation indicates that the SRM module is essential for transforming asynchronous multimodal signals into sparse and temporally precise event streams. In particular, the input response kernel $\varepsilon(t)$ and the self-spiking response kernel $\eta(t)$ form a sparse causal filtering mechanism, enabling the model to effectively capture temporal structures among cross-modal events.

Second, the liquid dynamics module is important for modeling continuous-time dependencies across events. When liquid-state propagation is removed, the model degenerates into a local event-response system, with MAE increasing to 0.341 and CAS and TC decreasing to 0.691 and 0.602, respectively. This demonstrates that the liquid layer can preserve inter-event dependencies and contextual integration through continuous-time state evolution, thereby enhancing global temporal consistency and historical state memory.

Third, the trace-variable propagation mechanism improves the stability of state evolution and computational efficiency. Removing this mechanism increases MAE to 0.329, decreases TC to 0.614, and makes state propagation less stable. Trace variables accumulate historical event responses as exponentially decaying temporal convolutions. This allows the model to retain historical information across sparse event intervals. They also avoid recomputing the full matrix exponential at every event time, reducing computational complexity while preserving analytical accuracy.

In addition, the hierarchical alignment objectives make important contributions to cross-modal temporal consistency modeling. Using only the spike-level loss lets the model learn local event correspondences but not the global temporal structure. This yields an MAE of 0.308 and a TC of 0.623. Removing the trajectory-level regularization term increases MAE from 0.291 to 0.295 and decreases TC from 0.658 to 0.634. This shows that trajectory-level consistency is important for maintaining the global alignment path over long temporal windows. The complete model jointly optimizes event-level, trajectory-level, and state-level constraints. This establishes multi-scale temporal consistency and substantially improves cross-modal alignment accuracy and temporal stability.

In summary, SRM encoding, liquid dynamics propagation, and the trace-variable mechanism jointly constitute the core structure of SynNeura. SRM encoding ensures temporal sparsity and event sensitivity in the input representation, liquid dynamics provides continuous-time state memory, and the trace-variable mechanism enhances the preservation of historical information and the efficiency of its propagation. Furthermore, the hierarchical alignment objective constrains the model with respect to local event matching, global trajectory consistency, and state stability. Therefore, SynNeura achieves a favorable balance among performance, stability, and interpretability in weakly supervised multimodal temporal alignment, fully validating its systematic advantages.

Fig. 11 examines robustness to partial spike omission by comparing firing-rate trajectories under clean versus corrupted inputs. The x-axis denotes time (s), and the y-axis denotes firing rate; the black dashed curve corresponds to the original signal, whereas the orange solid curve is obtained after removing a fraction of spikes.

The two trajectories remain highly similar in overall shape and peak timing, with only minor deviations in amplitude and local transition smoothness. Notably, during the 6–8s interval, the corrupted trajectory preserves the rhythmic pattern of the original one, showing slightly smoother peaks but no evident phase shift. This indicates that the dominant temporal structure is retained even when event information is partially missing.

From a modeling perspective, these observations provide qualitative evidence that SynNeura maintains temporal stability under input degradation. By combining event-triggered updates with continuous-time state evolution, the model can preserve rate-level temporal regularities and mitigate the impact of moderate spike loss. Overall, Fig. 11 supports that SynNeura yields robust temporal encoding in

Table 4
Ablation results of key components in SynNeura.

Model Vaariants	SRM Front-End	Liquid Solver	Alignment Loss Hierarchy	Trace-Varable Propagation	MAE↓	CAS↑	TC↑	EP/ER ↑
(a) w/o SRM	×	✓ (MatrixExp.)	✓ (Full)	✓	0.356	0.684	0.594	0.823 / 0.754
(b) w/o Liquid Dynamics	✓	×	✓	✓	0.341	0.691	0.602	0.835 / 0.768
(c) w/o Trace Variable	✓	✓	✓	×	0.329	0.701	0.614	0.847 / 0.773
(d) Spike-only Loss	✓	✓	× (Spike only)	✓	0.308	0.703	0.623	0.851 / 0.780
(e) w/o Trajectory Reg.	✓	✓	✓ (No_trajectory)	✓	0.295	0.708	0.634	0.864 / 0.796
(f) Full SynNeura	✓	✓ (Piecewise Exact)	✓ (Full)	✓	0.291	0.713	0.658	0.889 / 0.824

Table 5
Hyperparameter sensitivity and loss-term ablation analysis on the AVE dataset.

Hyperparameter Setting	Description	MAE ↓	TC ↑	CAS ↑
Default	$\Delta t = 0.5$, $\tau_c = 0.6$, $\lambda_1 = 0.5$, $\lambda_2 = 0.1$	0.291 ± 0.009	0.641 ± 0.017	0.713 ± 0.015
$\Delta t = 0.25$	Decreased temporal neighborhood window	0.306 ± 0.013	0.622 ± 0.021	0.694 ± 0.019
$\Delta t = 1.0$	Increased temporal neighborhood window	0.314 ± 0.015	0.615 ± 0.023	0.681 ± 0.021
$\tau_c = 0.4$	Lower similarity threshold	0.301 ± 0.012	0.627 ± 0.019	0.699 ± 0.018
$\tau_c = 0.8$	Higher similarity threshold	0.309 ± 0.016	0.620 ± 0.022	0.688 ± 0.020
$\lambda_1 = 0.25$	Lower trajectory-loss weight	0.299 ± 0.011	0.631 ± 0.018	0.701 ± 0.017
$\lambda_1 = 1.0$	Higher trajectory-loss weight	0.303 ± 0.014	0.626 ± 0.020	0.697 ± 0.022
$\lambda_2 = 0.05$	Lower state-regularization weight	0.297 ± 0.010	0.633 ± 0.021	0.704 ± 0.016
$\lambda_2 = 0.2$	Higher state-regularization weight	0.300 ± 0.013	0.629 ± 0.017	0.700 ± 0.019
w/o trajectory loss	$\lambda_1 = 0$, $\lambda_2 = 0.1$	0.317 ± 0.018	0.604 ± 0.026	0.676 ± 0.024
w/o state regularization	$\lambda_1 = 0.5$, $\lambda_2 = 0$	0.304 ± 0.014	0.618 ± 0.024	0.692 ± 0.021

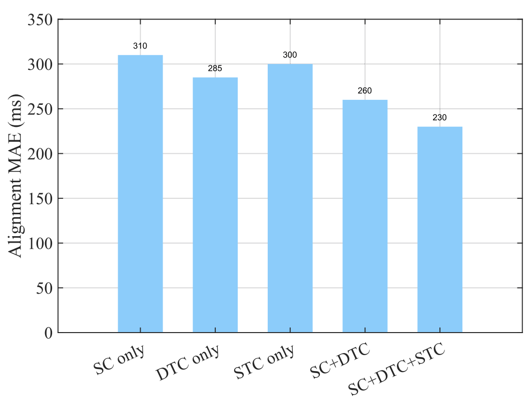


Fig. 12. Ablation study of loss components.

the presence of incomplete event streams, which is desirable for multimodal alignment under noisy or uncertain inputs.

Fig. 12 illustrates the effect of different loss combinations on cross-modal temporal alignment performance. The x-axis lists five configurations: SC only, DTC only, STC only, SC + DTC, and SC + DTC + STC. The y-axis reports the Alignment MAE (ms), with lower values indicating better alignment accuracy.

As additional objectives are introduced, MAE decreases from around 300 ms under single-loss training to below 200 ms when all three losses are jointly optimized, indicating that multi-objective learning improves temporal precision and cross-modal consistency. Among single-loss settings, SC-only yields the most significant error, suggesting that soft coupling alone provides limited temporal-structure constraints. DTC-only offers a moderate improvement by enforcing trajectory-level temporal consistency, while STC-only further reduces MAE, highlighting the benefit of jointly constraining spatio-temporal dynamics.

Combining objectives leads to clear synergy: SC + DTC substantially reduces MAE, and SC + DTC + STC achieves the best overall performance (MAE<200ms), suggesting that STC further enhances coordination within a shared spatio-temporal dynamic subspace. Overall, these results emphasize the importance of complementary, hierarchical losses for robust multimodal temporal alignment and provide empirical support for the proposed dynamic subspace consistency perspective.

4.5. Hyperparameter sensitivity analysis

To evaluate the robustness of SynNeura to key hyperparameter settings and the reproducibility of the experimental results, we further conduct a hyperparameter sensitivity analysis. Specifically, we select the temporal proximity threshold Δt, the similarity threshold τ_c, the trajectory loss weight λ_1, the state regularization weight λ_2, the InfoNCE temperature coefficient τ, and the Soft-DTW smoothing parameter γ as the analysis targets. In each experiment, only one hyperparameter is varied, while all others are kept at their default settings to examine its effect on model performance independently.

As shown in Table 5, SynNeura maintains relatively stable performance within a reasonable range of hyperparameter variations, indicating that the model does not rely on an extremely narrow hyperparameter tuning interval. Under the default setting, the model achieves an MAE of 0.291 ± 0.009, a TC of 0.641 ± 0.017, and a CAS of 0.713 ± 0.015 on the AVE dataset, outperforming the other perturbed hyperparameter configurations.

For the temporal proximity threshold Δt, model performance decreases when it is reduced from the default value of 0.5s to 0.25s, because a narrower window may exclude valid cross-modal event pairs and thereby weaken the event-level alignment signal. Conversely, when Δt is increased to 1.0s, performance also declines, as a wider window introduces more noisy candidate events and reduces the quality of cross-modal matching. Thus, a moderate temporal neighborhood constraint balances effective event coverage with noise suppression in cross-modal matching.

For the similarity threshold τ_c, the default value of 0.6 yields favorable results. When τ_c is set to 0.4, the lower threshold may introduce event pairs with weak semantic relevance. When τ_c is set to 0.8, the higher threshold may filter out some potentially valid positive pairs. Therefore, a moderate similarity threshold better balances positive-pair reliability and sample coverage.

In addition, the trajectory loss weight λ_1 and the state regularization weight λ_2 also have important effects on model performance. When λ_1 deviates from its default value or when the trajectory loss is removed, model performance degrades noticeably, indicating that trajectory-level temporal consistency constraints play a crucial role in guiding cross-modal state evolution. Similarly, an appropriate λ_2 helps stabilize the continuous-time evolution of liquid hidden states. When state regularization is too weak or removed, the stability of state evolution deteriorates, whereas overly strong regularization may suppress useful dynamic variations.

Overall, Δt and τ_c mainly affect the quality of cross-modal candidate event construction, λ_1 controls the strength of trajectory-level temporal consistency constraints, and λ_2 influences the numerical stability of liquid-state dynamics. These results validate the effectiveness of SynNeura in event matching, semantic filtering, trajectory constraint modeling, and state stabilization, while also demonstrating its favorable robustness to hyperparameters and interpretability.

5. Conclusion

This paper presents SynNeura, an event-driven liquid-spiking neural dynamics framework for weakly supervised multimodal temporal alignment. SynNeura formulates multimodal alignment as a continuous-time state-evolution process. Asynchronous multimodal event streams are efficiently modeled using closed-form analytical propagation between events, with sparse nonlinear updates upon event arrival. This design reduces the reliance on dense sampling and fixed temporal grids. To address the lack of precise alignment annotations under weak supervision, SynNeura introduces hierarchical alignment objectives at the event, trajectory, and state levels. These enforce cross-modal temporal consistency, combining local event matching, global dynamic coherence, and continuous-state stability.

Extensive experiments across multiple benchmarks, together with ablation studies and complexity analyses, demonstrate that SynNeura outperforms representative baselines in alignment accuracy, cross-modal temporal consistency, and inference efficiency, and validate the effectiveness of its core components. Since its dominant computational cost scales with the number of events rather than the length of the temporal grid, SynNeura exhibits favorable scalability and strong potential for online inference in long-sequence, low-sampling-rate, and event-sparse scenarios.

Nevertheless, SynNeura still has certain limitations. The piecewise-constant approximation between events may lead to local information loss when extremely high-frequency dynamics, persistent oscillations, or fine-grained variations are insufficiently captured by event triggers. Future work will explore adaptive event thresholds, local higher-order approximations, and uncertainty estimation mechanisms to enhance the model's ability to represent rapid dynamics and improve alignment reliability in complex scenarios.

CRediT authorship contribution statement

Yuping Zhang: Writing – original draft, Validation, Data curation, Conceptualization; **Yan Liu:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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